

The Hierarchical Latent Workspace Model

Root-Preserving Variable-Depth Expertise for Parallel Diffusion Reasoning

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Preprint · 4 September 2026

Abstract

Most language models pair a fixed computational graph with a single public generation trajectory. Sparse mixtures of experts vary *which* parameters run, and diffusion language models revise blocks in parallel, but neither mechanism alone provides what deliberate problem solving needs: task-wide framing, specialist work in isolated workspaces, explicit verification, and a guarded decision about what to publish. We propose the *Hierarchical Latent Workspace Model* (HLWM): one model whose computation is a dynamically selected subgraph of a fixed candidate expert graph, whose reasoning happens in state-isolated parallel lanes refined by categorical diffusion, and whose only public output is an exact token snapshot that has passed calibrated commitment, risk, and verification heads. A permanent shared substrate and an always-executed root expert make specialization additive rather than substitutive: unselected specialists simply do not run, and the model has no single fixed active-parameter count.

We report ten completed preregistered prototype experiments on a frozen Qwen3-0.6B substrate with trainable sidecars, each gated before the next was authorized. All three proposed mechanisms have now been falsified at this scale, on the record. The routed expert graph passed its preregistered load gates on one seed of two, at six experts twice and at two experts once, but its causal-liveness gate failed on ten of ten seed-runs and the sibling seed collapsed to a single expert under three successive configurations, including the minimum viable count of two; the preregistered reduction rule retired it, at a measured cost of 0.19 to 0.28 of probe content accuracy, which showed the expert bank had been carrying content capacity while failing every routing gate. Verified fan-in failed next: its selector was near-oracle, on one seed reaching its own any-valid oracle exactly, yet it lost to self-consistency by roughly thirty points, because a single mis-specified token identifier, a beginning-of-sequence id that the pinned tokenizer aliases to end-of-text, had knocked the workspace generation channel into the wrong register. A dedicated repair experiment fixed that channel under a frozen preregistration and then measured the third mechanism directly. The repair is confirmed on both seeds: zero turn-scaffold emissions in 640 audited rows, and workspace-conditioned decoding within 0.013 of causal accuracy where the fan-in experiment had a thirty-point gap. The mechanism is not: ablating the read-out's 34 memory positions changes graded accuracy by 0.000 on one seed and improves it by 0.022 on the other, and the full prefix is bounded by parity itself, since the causal arm is the workspace arm with the prefix removed. Parity was reached by irrelevance: both channels read the same full context, the latent was redundant by construction, and the model behaved accordingly. A final necessity experiment removed the redundancy: on masked rows the answer path was physically denied the operative premise, the workspace alone saw it, and the model trained under that asymmetry. The instrument was airtight, zero leaked rows in 198 masked audits, a causal floor of exactly 0.000, and a trained attribution control, and the reading was zero on both seeds: masked accuracy 0.000 on 99 rows per seed, while a linear probe still decodes roughly 0.30 of the withheld premise from the latent state. The information reaches the latent and generation cannot use it; the preregistered terminal rung retires the generative workspace mechanism at this scale.

What survives is narrower than the architecture and is reported as such. The split-conformal publication rule delivered its coverage floor where the earlier midpoint rule had strangled, and its operating points were flawless on both seeds: 138 published answers, 138 correct, with total refusal on the one audited domain where the model is incompetent. It lost its preregistered risk-coverage comparison to matched mean-log-probability abstention once; under the unchanged comparison in the final experiment, the calibrated

heads beat that baseline directionally on both seeds for the first time in the program, at paired-bootstrap win fractions of 0.746 and 0.792 against the preregistered 0.80 bar, with operating points again flawless (77 of 77 published answers correct on one seed). The claim remains unestablished and is reported as a near-miss with a power analysis, not a victory. Selection was declared void under its own headroom gate: the audit distribution proved bimodal, with no exactly-one-correct rows in the medium stratum on either seed. The program therefore closes zero for three on mechanisms. Its deliverables are the negative results themselves, a failure taxonomy in which every observed pathology has a named detector and a repair or a retirement, and the evaluation machinery: semantic graders, generated-emission calibration, split-conformal thresholds, domain-stratified audits, and gates declared before every run. Nothing here is a production-readiness or model-superiority claim; the decisive matched plain-LoRA control was never authorized by any gate pass. The final experiment establishes that structural necessity is necessary but not sufficient: the latent carried the information and the decoder still could not spend it. Any continuation must additionally supervise the latent channel densely, with token-level reconstruction or distillation targets through the latent itself, the one ingredient shared by every published working system in this class and absent from every experiment here.

1. Introduction

Autoregressive generation has a useful discipline: every new token is conditioned on a stable prefix. Its weakness is equally structural. Once an early token is sampled, subsequent computation inherits that choice until a new sequence is generated. Diffusion language models relax this local commitment by revising a block bidirectionally, but parallel token revision is not the same as parallel problem solving. Sparse mixtures of experts provide conditional capacity, but several experts transforming one shared hidden state do not themselves preserve incompatible hypotheses or assign state-isolated scopes of work.

The proposed HLWM combines three ideas into one coherent computation.

1. A *root-preserving expert graph* provides variable-depth conditional expertise rather than fixed top- k routing. A budgeted dispatcher activates only the nodes predicted useful for the current route window; a permanent shared substrate and an always-executed root expert run in every functional stage.
2. A *multi-timescale latent workspace* dispatches distinct private briefs to state-isolated tensor lanes and repeatedly applies one shared transition: fast states revise candidate work at every microstep, private slow states consolidate progress at window boundaries, and the global workspace updates only after verified fan-in at synchronization barriers.
3. *Diffusion refinement with guarded commitment* permits revision before any candidate becomes public. Commitment freezes an exact token snapshot; calibrated candidate-level heads decide whether that snapshot is published or a fixed safe fallback is returned. Private canvases never publish directly.

The architecture is one neural model: experts are parameter blocks, lanes are batched tensor paths, recurrence increases executed depth without allocating a new block per step, verification is an internal mode, and publication is an exact-snapshot decoding decision. It is not a collection of agents exchanging natural-language messages.

The architecture in one picture. Concretely, an HLWM is a frozen pretrained decoder wrapped by three trainable subsystems (Figure 1). A *workspace* reads the full input and produces a small set of private latent states, the model’s own hidden representations recirculated as continuous thoughts rather than sampled tokens. An *interface* makes those states visible to the decoder: latent thoughts sit in the input sequence, and their projections are appended to every attention layer’s keys and values through learned per-layer, per-head gates, so the decoder can consult the workspace at any depth without the workspace ever emitting a public token. A *publication layer* scores every candidate answer with calibrated commitment, risk, and verification heads and either publishes one exact token snapshot or returns a safe abstention; coverage is controlled by a split-conformal threshold rather than a tuned constant. Specialist capacity enters through low-rank experts on the decoder’s feed-forward projections, routed deterministically by task family so that routing cannot collapse by construction. The current instantiation trains the workspace channel with dense supervision on every latent position, distilled from gold reasoning traces (Section 9.12); the experiments of Section 9 document why every weaker training signal failed first.

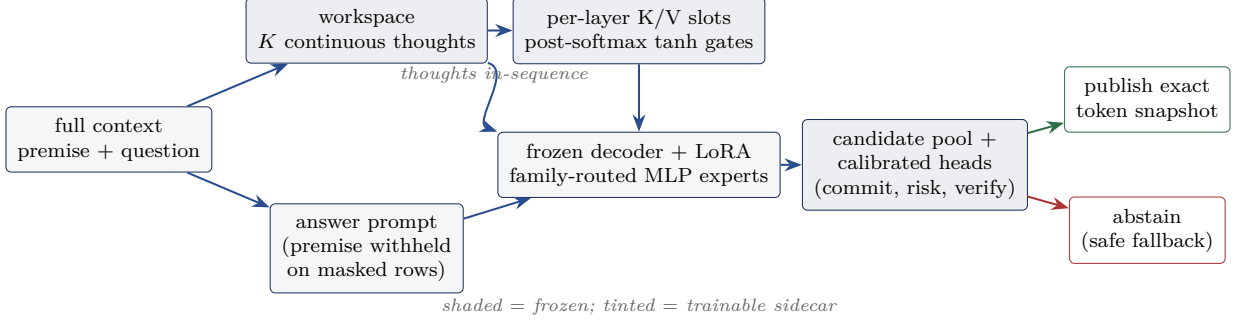


Figure 1: The hierarchical latent workspace model. A frozen decoder is wrapped by three trainable subsystems: a workspace that recirculates the model’s own hidden states as continuous thoughts, a gated per-layer key–value interface that makes the workspace visible at every depth, and a calibrated publication layer that either publishes one exact token snapshot or abstains. On masked rows the answer prompt is physically denied the premise, so the workspace channel is the only route from evidence to answer; structural necessity is a property of the data, not a regularizer.

The central research question is:

Can a root-preserving, variable-depth expert graph coordinate isolated parallel reasoning and private block revision while improving verified answer quality at matched active computation?

The contribution is not simply more experts or more thinking. It is a three-axis allocation rule: connected expert-graph depth, per-lane recurrent depth, and macrocycle count, all charged against one declared budget. A general question may execute only the shared substrate and root for a few recursive calls. A Python debugging lane may begin with a broad software route, consolidate what it has learned, descend through Python into debugging and testing specialists, and halt independently when another update has no measured value. A multidisciplinary task may activate several routes and several lanes concurrently. In every case, the shared substrate and root remain structurally active, their functional contribution is measured through interventions and anchor tasks, and one budget limits expert-graph depth, private recursive depth, and macrocycle depth.

Contributions. This paper makes four contributions.

1. **An architecture specification** for hierarchical latent workspace models: a root-preserving expert graph, a multi-timescale isolated-lane workspace with diffusion refinement, and calibrated atomic publication, all charged against a single declared compute budget (Sections 2–7).
2. **A preregistered evaluation methodology** for architecture claims at small scale: gate batteries, margins, and binding decision ladders frozen before every run; killer baselines inside every audit; semantic graders and split-conformal publication thresholds fitted only on generated held-out emissions (Section 8).
3. **A complete experimental record** of ten gated experiments on a frozen 0.6B substrate (Section 9). All three proposed mechanisms are falsified at this scale, each with a mechanism-level diagnosis rather than a bare null: learned routing collapses whenever the assignment is learnable; candidate selection is void without measured difficulty headroom; and the latent workspace channel carries probe-decodable content that generation cannot use, first through redundancy and finally under structural necessity (masked-premise transfer 0.000 on both seeds while a linear probe recovers a third of the withheld content). The one surviving positive thread is calibrated selective prediction, whose operating points replicate flawlessly and whose advantage over free baselines is directionally consistent but below its preregistered significance bar.
4. **A transferable failure taxonomy** for bolt-on reasoning structures: every observed pathology (scale-invariant gates, token-id aliasing on no-BOS bases, padded harvest surfaces, degenerate calibration thresholds, schedule–curriculum aliasing) is named, given a detector, and paired with the instrument correction that exposed it (Section 10).

Reading guide. Sections 2–7 specify the architecture and training proposal; Section 8 the protocol; Section 9 the ten experiments; Section 10 failure modes and decision rules; Section 11 limitations; Section 12 related work. The proposed mechanisms are retained in full as proposal plus negative evidence; a reader who wants only the outcomes should read Sections 9.9, 9.10, 9.11 and 13. Artifact provenance is in Appendix A.

2. Design Specification

2.1 Operational commitments and testable assumptions

The proposal separates mechanisms that define the model from effects that experiments must establish.

- I1. Permanent general substrate.** Shared embeddings, attention, normalization, and language-output machinery execute in every functional stage.
- I2. Always-executed root.** A root/generalist expert cannot be removed by graph routing. No fixed mixture percentage is claimed to preserve capability; preservation is evaluated behaviorally through preregistered non-inferiority and intervention tests.
- I3. Variable-depth graph activation.** The router chooses stop or expansion decisions over a fixed candidate expert directed acyclic graph (DAG). It does not activate a fixed number of experts or traverse a fixed depth.
- I4. Dormancy by default.** Specialists outside the selected stage-and-lane subgraph do not execute. They may remain resident in memory, but incur no expert forward pass or gradient to their expert parameters for that invocation; the router may still receive exploration or regularization gradients.
- I5. Private parallel work.** Solver lanes cannot read one another’s private states. Cross-lane exchange occurs only through identified summaries, verification records, and synthesis.
- I6. Shared recursive transition.** The same root-preserving transition kernel is reused across lanes, functional modes, and recursive depth. Recurrence adds executed computation without allocating depth-specific transformer blocks.
- I7. Barrier-mediated global state.** Private lanes update concurrently against a frozen global workspace. Only structured summaries and verification records cross a synchronization barrier; hidden lane states never enter the global workspace directly.
- I8. Atomic publication.** Commitment freezes an exact token-ID snapshot. Only those buffered IDs may be appended, after which they are causally re-encoded to extend the public key–value cache. Solver canvases never publish directly.

These invariants distinguish the proposal from a conventional fixed-width mixture of experts (MoE), an external tree-search controller, and a multi-agent workflow.

2.2 Two hierarchies and model state

HLWM contains two different structures. The *functional hierarchy* describes the flow of work:

framing $\rightarrow$ decomposition $\rightarrow$ solving $\rightarrow$ verification $\rightarrow$ synthesis $\rightarrow$ commitment.

The *parameter graph* is a fixed typed candidate DAG $\mathcal{G}_{E,\text{cand}} = (\mathcal{V}, \mathcal{E})$ chosen before a training run:

root $\rightarrow$ domain $\rightarrow$ capability $\rightarrow$ technology/method $\rightarrow$ operation.

Training learns the expert parameters and a conditional routing policy over its permitted edges; version one does not add or delete graph edges. The graph need not be a strict tree: a testing specialist, for example, may have several admissible parents. Semantic node names are initialization priors and diagnostic labels, not guarantees about the learned function.

For output block b and macrocycle r , the model maintains the state in Table 1.

Table 1: Principal state variables. Private state is never appended to committed context unless synthesis and commitment accept it.

Symbol	Meaning
C_b	Causally encoded user input and previously committed blocks; fixed while block b is refined.
$g_{b,r}$	Global workspace containing the objective, hard constraints, unresolved dependencies, and verified findings.
$\mathcal{A}_{b,r,q}^{m,j}$	Active root-connected expert subgraph for stage m , lane j , and route window q ; shared stages omit j .
$B_{b,r}^j$	Learned brief for solver lane j : scope, assumptions, deliverable, dependencies, and failure tests.
$w_{b,r,q}^j$	Private slow lane state containing its consolidated plan, unresolved dependencies, and progress after route window q .
$z_{b,r,q,s}^j, D_{b,r,q,s}^j$	Private fast latent state and categorical canvas at microstep s inside route window q .
$A_{b,r}^j$	Identified branch summary containing claims, dependencies, route provenance, and uncertainty.
$V_{b,r}^k$	Structured verification record for assigned claims or candidates.
$Y_{b,r}$	Separately refined private synthesis block; the only block eligible for publication.

3. Dynamic Root-Preserving Expert Graph

3.1 Stage-progressive, budgeted graph traversal

Let the fixed candidate DAG be $\mathcal{G}_{E,\text{cand}} = (\mathcal{V}, \mathcal{E})$ with root node v_0 . Immediately before stage m , the dispatcher observes every result that is legitimately available at that point:

$$\xi_{b,r}^{m,j} = (C_b, g_{b,r}, e_m, e_j, \text{stage inputs, verified history, remaining budget}),$$

where e_m and e_j are stage and lane embeddings. The router predicts a logit $\ell_v(\xi)$ for every specialist reachable under the current ancestry mask. A horizon head also predicts ordered continuation logits $r_{m,j,1:N_{\max}^m}(\xi)$; accepting increment n requires accepting every earlier increment, so the requested fast-update cap $N_{\text{cap}}^{m,j}$ is a prefix rather than an unconstrained integer. The stage schedule deterministically expands that cap into $N_{\Phi}^{m,j}$ shared-kernel invocations, including every required private consolidation. For route j in stage m , binary execution gates and this complete scheduled count satisfy

$$a_{v_0}^{m,j} = 1, \tag{1}$$

$$a_v^{m,j} \leq \sum_{p \in \text{Pa}(v)} a_p^{m,j} \quad (v \neq v_0), \tag{2}$$

$$\text{EC}(\mathcal{A}^{m,j}, N_{\Phi}^{m,j}) = c_{\text{route}} + c_{\text{aux}}^{m,j} + N_{\Phi}^{m,j} \left(c_{\text{shared}} + c_0 + \sum_{v \in \mathcal{A}^{m,j} \setminus \{v_0\}} c_v \right). \tag{3}$$

Here EC denotes execution cost. Every active specialist has a recursively active path to the root, activating no child is the specialist stop action, and repeated use of an expert is charged repeatedly. $N_{\Phi}^{m,j}$ includes fast transitions and scheduled private consolidations; $c_{\text{aux}}^{m,j}$ covers declared heads, routing, summary extraction, and other non-kernel work assigned to that reservation. Version one learns the gates and expert parameters, not the candidate topology.

One atomic block ledger first escrows a private causal-fallback reserve and exposes only the remaining balance to macrocycles. Before any parallel interval, every unfinished lane submits route scores and a requested fast-update cap. A joint allocator chooses $\{(\mathcal{A}^{m,j}, N_{\text{cap}}^{m,j})\}_{j \in \mathcal{J}_q}$ and expands each cap into its complete scheduled count $N_{\Phi}^{m,j}$ under

$$\sum_{j \in \mathcal{J}_q} \text{EC}(\mathcal{A}^{m,j}, N_{\Phi}^{m,j}) + B_{\text{downstream}} \leq B_{\text{remain}}, \tag{4}$$

where $B_{\text{downstream}}$ reserves declared minimum cost for routing and control, branch summaries, verification, the global slow update, synthesis, the root-preserving commitment transition, risk and expected-value-of-information (EVI) scoring, and causal cache extension. The fallback is excluded here because one fallback escrow, covering control, causal generation, commitment-state construction, risk scoring, and

possible cache extension, was created exactly once at block-ledger initialization; B_{remain} is the spendable balance after that escrow. Every learned forward pass and every non-negligible routing, checking, or control operation must draw from a named reservation before executing and atomically settle its measured cost afterward. Constant-time ledger comparisons are not learned model computation. If the minimum per-lane reservations are infeasible, the dispatcher reduces the lane set before execution; no lane may spend another lane’s reservation speculatively. Consequently, simultaneous initial routes, consolidations, refreshes, scoring, and publication cannot exceed the block budget.

During training, each discrete node bid uses a straight-through Binary-Concrete estimator. The horizon request uses relaxed prefix gates, so its previously implicit estimator is explicit. For independent $u_v, u_n \sim \text{Uniform}(0, 1)$, let $g(u) = \log u - \log(1 - u)$ and define

$$\tilde{a}_v = \sigma\left(\frac{\ell_v + g(u_v)}{\tau_a}\right), \quad (5)$$

$$\tilde{c}_{m,j,n} = \prod_{i=1}^n \sigma\left(\frac{r_{m,j,i} + g(u_i)}{\tau_N}\right), \quad \tilde{N}_{\text{cap}}^{m,j} = \sum_{n=1}^{N_{\text{max}}^m} \tilde{c}_{m,j,n}, \quad (6)$$

$$(\mathbf{a}^{\text{hard}}, \mathbf{N}_{\text{cap}}^{\text{hard}}) = \Pi_{B_{\text{remain}}}(\tilde{\mathbf{a}}, \tilde{\mathbf{c}}; \mathcal{G}_{E,\text{cand}}), \quad (7)$$

$$\mathbf{a}^{\text{ST}} = \text{stopgrad}(\mathbf{a}^{\text{hard}} - \tilde{\mathbf{a}}) + \tilde{\mathbf{a}}, \quad (8)$$

$$\mathbf{N}_{\text{cap}}^{\text{ST}} = \text{stopgrad}(\mathbf{N}_{\text{cap}}^{\text{hard}} - \tilde{N}_{\text{cap}}) + \tilde{N}_{\text{cap}}. \quad (9)$$

The deterministic projection $\Pi_{B_{\text{remain}}}$ first reserves the minimum complete schedule for every admitted lane, then adds feasible transition calls and nodes, together with missing ancestors, in marginal score-to-execution-cost order while enforcing equation (4). A call increment is admitted only after its prefix and any required consolidation call; stable lane and node identifiers break ties. The forward pass executes only the hard route and hard horizon, so an inactive expert receives neither an expert forward pass nor an expert-parameter gradient. The relaxed terms supply biased, low-variance router and horizon gradients; score-function estimators are required controls. At inference, noise is removed and the same joint projection is applied to node and ordered continuation logits. This is a declared version-one estimator, not a claim that it is optimal [1].

Framing is routed from committed context and the previous frame; decomposition is routed after the new frame exists; solver paths are conditioned on learned briefs; verifier paths are conditioned on resulting claims; and synthesis is routed after verification findings exist. Within a recurrent stage, a route is fixed for one bounded window of at most K fast updates. At the next boundary, each unfinished lane may form a refresh bid from only its own consolidated private state; the joint allocator arbitrates all bids against the shared ledger. This preserves state isolation and coherent specialist work inside a window while allowing broad early routes to descend toward newly discovered specialist operations without exceeding the macrocycle budget.

In plain terms: the model bids for computation the way a project bids against a budget. Each lane proposes how deep it wants to specialize and how many refinement steps it wants; a single ledger admits what fits, always keeping enough in reserve to verify, synthesize, and publish safely.

This produces a distribution of active computation rather than one advertised number. For one stage and lane,

$$P_{\text{active}}^{m,j}(x) = P_{\text{trunk}} + P_0 + \sum_{v \in \mathcal{A}^{m,j}(x) \setminus \{v_0\}} P_v. \quad (10)$$

A proper report gives minimum, mean, 95th percentile, maximum, and per-stage distributions of active parameters and FLOPs. An eight-billion-active label is not an architectural constant of HLWM.

3.2 Always-executed root and behavioral retention

Simply executing the root does not establish that it contributes useful general capability. Conversely, a large residual coefficient does not measure retained language ability: later layers, vector cancellation, and the shared trunk all intervene. HLWM therefore separates *structural root presence* from *behavioral preservation*. For expert v , let RMS denote root-mean-square magnitude over hidden dimensions and

define a normalized output

$$\hat{E}_v(h) = \frac{E_v(h)}{\text{RMS}(E_v(h)) + \epsilon_{\text{num}}}. \quad (11)$$

Let T_θ be the permanent shared substrate, let $\gamma_0 = \text{softplus}(\eta_0)$ and $\gamma_s = \text{softplus}(\eta_s)$ be learned layerwise scales that do not depend on the task route, and let $s(x, m, j) = \text{softplus}(a_s)$ be a nonnegative route-conditioned specialist strength. A connected graph is not executed as a flat weighted sum. It performs one topological downward sweep followed by a reverse-topological upward sweep. With α_{pv} normalized over the active parents of v , β_{vc} assigning each child message across its active parents, and LN denoting layer normalization, define

$$h_{\text{shared}} = T_\theta(h), \quad r_0 = \hat{E}_0(h_{\text{shared}}), \quad (12)$$

$$r_v = \hat{E}_v \left(\text{LN} \left[h_{\text{shared}} + \gamma_0 W_0 r_0 + \sum_{p \in \text{Pa}_{\mathcal{A}}(v)} \alpha_{pv} U_{pv} r_p \right] \right), \quad v \in \mathcal{A} \setminus \{v_0\}, \quad (13)$$

$$u_v = r_v + \sum_{c \in \text{Ch}_{\mathcal{A}}(v)} \beta_{vc} R_{vc} u_c, \quad v \in \mathcal{A} \text{ in reverse order}, \quad (14)$$

$$h' = h_{\text{shared}} + \gamma_0 W_0 r_0 + s(x, m, j) \gamma_s W_s (u_0 - r_0). \quad (15)$$

For a DAG, $\sum_{p \in \text{Pa}_{\mathcal{A}}(c)} \beta_{pc} = 1$ prevents a multiply parented child from being counted once per parent. If no specialist is active, $u_0 = r_0$ and the specialist residual is exactly zero. Active siblings at the same graph depth execute concurrently; only ancestor–descendant dependencies impose sequential graph depth. Thus adding a deeper node changes both the context it receives and the evidence returned toward the root, rather than merely adding another flat expert output.

The root has no router-controlled gate: its message conditions every active specialist, remains an explicit residual in the final update, and is recomputed on every recursive invocation. Specialist strength may become large on narrow work, but no scalar in equation (15) is interpreted as a percentage of intelligence, fluency, or reasoning. Layerwise root and specialist residual norms, edge-message norms, directional alignment, and ablation effects are reported as diagnostics. Held-out behavior, rather than γ_0 , determines whether the root remains functionally useful.

Functional preservation is defined on preregistered anchor families $\{\mathcal{D}_k\}_{k=1}^K$ covering language modeling, instruction adherence, discourse coherence, global constraints, and other claimed general capabilities. Let p_{ref} be a frozen competent reference, p_{root} the current shared-plus-root path with specialists disabled, and p_{full} the routed model. For a loss $\mathcal{L}_k$ whose lower value is better, define

$$\Delta_k^{\text{root}} = \mathcal{L}_k(p_{\text{root}}; \mathcal{D}_k) - \mathcal{L}_k(p_{\text{ref}}; \mathcal{D}_k), \quad (16)$$

$$\Delta_k^{\text{route}} = \mathcal{L}_k(p_{\text{full}}; \mathcal{D}_k) - \mathcal{L}_k(p_{\text{root}}; \mathcal{D}_k). \quad (17)$$

Training targets the constrained problem

$$\min_{\theta} \mathcal{L}_{\text{task}} + \lambda_C \mathbb{E}[\text{Cost}(x)] \quad \text{subject to} \quad \Delta_k^{\text{root}} \leq \epsilon_k^{\text{root}}, \quad \Delta_k^{\text{route}} \leq \epsilon_k^{\text{route}} \quad \forall k. \quad (18)$$

The tolerances ϵ_k must be declared before final evaluation and justified by measurement uncertainty or application requirements. They are not universal constants. Dual optimization or an equivalent Lagrangian can optimize toward the constraints; held-out confidence bounds determine whether the constraints are actually satisfied. Counterfactual root ablation, specialist masking, and held-out anchor results determine whether the root is functionally useful; architectural presence alone is insufficient. Specialist value on domain d is measured by the counterfactual gain

$$G_{\text{spec}}(d) = \mathcal{R}_d(p_{\text{root}}) - \mathcal{R}_d(p_{\text{full}}), \quad (19)$$

for a preregistered risk $\mathcal{R}_d$ whose lower value is better. This tests whether specialists add capability rather than merely receive high routing scores.

Adaptive halting exposes several possible readout depths, so terminal-only retention is insufficient. For halt-eligible readouts $q \in \mathcal{Q}_{\text{halt}}$, define both the root-only and routed gaps

$$\Delta_{k,q}^{\text{root}} = \mathcal{L}_k(p_{\text{root}}^{(q)}; \mathcal{D}_k) - \mathcal{L}_k(p_{\text{ref}}; \mathcal{D}_k), \quad (20)$$

$$\Delta_{k,q}^{\text{route}} = \mathcal{L}_k(p_{\text{full}}^{(q)}; \mathcal{D}_k) - \mathcal{L}_k(p_{\text{root}}^{(q)}; \mathcal{D}_k), \quad (21)$$

$$\max_{q \in \mathcal{Q}_{\text{halt}}} \Delta_{k,q}^{\text{root}} \leq \epsilon_k^{\text{root}}, \quad \max_{q \in \mathcal{Q}_{\text{halt}}} \Delta_{k,q}^{\text{route}} \leq \epsilon_k^{\text{route}}. \quad (22)$$

Training may sample readout depths to estimate these constraints, but final evaluation tests both gaps at every declared halt-eligible depth. Root execution at every recursive call is structural; retained competence of both the root-only and routed paths at those calls remains a behavioral claim.

3.3 Illustrative activation, not a handwritten schedule

Figure 2 shows the intended qualitative behavior. A general prompt stops at the root. A Python debugging request expands a relevant connected subgraph; unrelated domains remain dormant. The dispatcher learns this behavior from outcomes and budget pressure. Training records do not prescribe hand-authored phase weights.

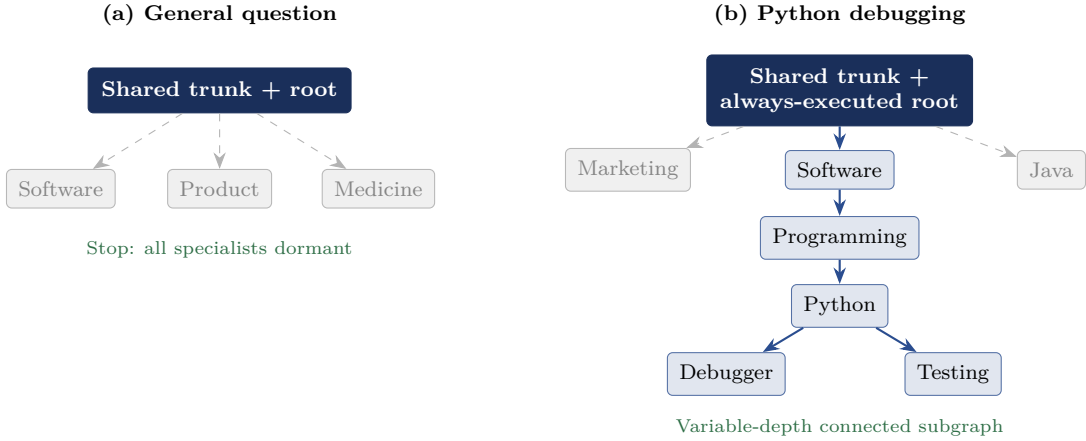


Figure 2: Task-dependent activation of the expert graph. Shaded nodes execute; gray nodes do not. The example illustrates expected behavior, not fixed semantic routes or manually supplied weights.

4. Parallel Latent Workspace

4.1 Adaptive lanes, distinct briefs, and isolation

The architecture provides at most J_{max} solver lanes, but the dispatcher activates a task-dependent subset $\mathcal{J}_{b,r}$. A simple prompt may use one root-only lane. A difficult debugging task may use implementation, test, and adversarial lanes. Joint decomposition first proposes a candidate lane set and its briefs, then one allocator admits a feasible subset and all initial schedules together:

$$\left(\tilde{\mathcal{J}}, \{B^j\}_{j \in \tilde{\mathcal{J}}} \right) = \text{BriefHead}(C, g, \{e_j\}, B_{\text{remain}}), \quad (23)$$

$$\left(\mathcal{J}, \{(\mathcal{A}_0^{m,j}, N_{\text{cap}}^{m,j})\}_{j \in \mathcal{J}} \right) = \text{JointAllocate}\left(\{\text{RouteBid}(C, g, B^j)\}_{j \in \tilde{\mathcal{J}}}, B_{\text{remain}}\right), \quad \mathcal{J} \subseteq \tilde{\mathcal{J}}. \quad (24)$$

Each brief specifies scope, allowed assumptions, expected deliverables, dependencies, and rejection tests. Branches may be complementary, competitive, or adversarial. Diversity is evaluated from verified informational contribution rather than route names or raw token disagreement. Let $\mathcal{R}(x)$ be the finite set of evaluator-defined requirements for task x , let C_j be the atomic claims recorded by lane j , and let $v(c) \in \{0, 1\}$ be an executable or independently adjudicated correctness label. With $C_j^+ = \{c \in C_j : v(c) = 1\}$, define

$$\text{Cov}(x) = \frac{1}{|\mathcal{R}(x)|} \sum_{r \in \mathcal{R}(x)} \mathbf{1}[\exists j, c \in C_j^+ : c \models r], \quad (25)$$

$$\text{Red}(x) = \frac{2}{J(J-1)} \sum_{i < j} \frac{|C_i^+ \cap C_j^+|}{|C_i^+ \cup C_j^+|}, \quad (26)$$

$$M_j(x) = U(Y_{\text{all}}, x) - U(Y_{\setminus j}, x). \quad (27)$$

Here $\sim$ is a preregistered semantic-equivalence rule. Redundancy is reported only for $J \geq 2$ and nonempty unions. To estimate M_j , synthesis is rerun without summary A^j while its route, compute budget, and sampling randomness are held fixed. Multiple lanes count as useful only when they improve verified coverage or utility without increasing verified error; disagreement alone is not productive diversity.

During solving, lane j may attend to C , g , B^j , w^j , z^j , and D^j , but not to another lane’s private state. The public inputs are read-only during the solver interval, and attention masks enforce

$$\frac{\partial H_{q,s}^j}{\partial w_{q'}^i} = 0, \quad \frac{\partial H_{q,s}^j}{\partial z_{q',s'}^i} = 0, \quad \frac{\partial H_{q,s}^j}{\partial D_{q',s'}^i} = 0 \quad (i \neq j) \quad (28)$$

within that interval. A synchronization barrier materializes every identified summary before verification or synthesis can read it. Shared parameters, common context, and related briefs can still correlate outputs; *state-isolated* is an information-flow claim, not statistical independence. After solving, the structured summary

$$A^j = \text{SummaryHead}(B^j, w^j, z^j, D^j, \mathcal{A}_{0:Q_j}^{\text{solve},j}) \quad (29)$$

records atomic claims, requirement links, dependencies, uncertainty, and route provenance for downstream verification and synthesis. This boundary preserves alternatives long enough to be useful without turning the model into communicating agents.

Because the transition parameters are shared across lanes, collapse is an explicit risk. Lanes are differentiated by jointly generated briefs, lane embeddings, independently keyed initialization noise, and potentially different routes. Hidden-state repulsion is not treated as evidence of useful diversity because it can reward different errors. The accepted criteria remain verified coverage, verified redundancy, and fixed-randomness marginal synthesis utility from equation (27).

4.2 Root-preserving multi-rate recurrence

The same private recurrence is used for stages $m \in \mathcal{M}_{\text{priv}} = \{\text{solve}, \text{verify}, \text{synthesize}\}$. Its stage input $B^{m,j}$ is respectively a solver brief, a claim assignment with rejection tests, or a verified integration contract; synthesis normally uses one lane, while solving and verification may use several. Every newly activated private lane receives a slow state, fast state, categorical canvas, and empty self-conditioning state. For readability, the following equations suppress m on recurrent states, but initialization, mode embeddings, output heads, validity boundaries, utilities, and maximum horizons remain stage-conditioned:

$$w_0^j = \text{Init}_w(C_b, g_{b,r}, B^{m,j}, e_m, e_j), \quad (30)$$

$$z_{0,0}^j = \text{Init}_z(C_b, g_{b,r}, B^{m,j}, w_0^j, e_m, e_j) + \sigma_z \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, I), \quad (31)$$

$$D_{0,0}^j \sim \text{Cat}(\nu)^{L_j}, \quad S_{0,0}^j = 0. \quad (32)$$

All recurrent roles reuse one transition kernel Φ_θ under functional-mode, stage, lane, timescale, and depth embeddings. Small mode-specific projections map state types to and from its common width; no transformer block is allocated to a particular lane or recursive step. At fast microstep s of route window q , lane j forms

$$\chi_{q,s}^j = \text{Pack}(C_b, g_{b,r}, B^{m,j}, w_q^j, z_{q,s}^j, D_{q,s}^j, S_{q,s}^j, e_m, e_j, e_q, e_s), \quad (33)$$

$$H_{q,s}^j = \Phi_\theta(\chi_{q,s}^j; \mathcal{A}_q^{m,j}). \quad (34)$$

Every invocation of Φ_θ applies the permanent shared substrate and the root-preserving graph operator in equation (15); only specialists in $\mathcal{A}_q^{m,j}$ are conditional. Reusing Φ_θ increases executed depth without increasing stored depth-specific parameters. This weight tying is a testable efficiency choice, not a guarantee of improved reasoning.

Flatten (q, s) to update index $n = qK + s$. A declared stage schedule $t_{q,s} = \tau_m(n)$ maps each executed update to its reverse categorical-diffusion timestep; targeted re-noising is represented by an explicit per-position timestep mask rather than an undefined reset of $t_{q,s}$.

The fast head updates the private latent and diffusion states and predicts the externally testable value

of another update:

$$(\bar{z}_{q,s+1}^j, u_{q,s}^j, P_{q,s}^j, \eta_{q,s}^j, \hat{d}_{q,s}^j) = \text{FastHead}(H_{q,s}^j), \quad (35)$$

$$z_{q,s+1}^j = (1 - \sigma(u_{q,s}^j)) \odot z_{q,s}^j + \sigma(u_{q,s}^j) \odot \bar{z}_{q,s+1}^j, \quad (36)$$

$$\bar{D}_{q,s+1}^j \sim \text{ReverseCat}(D_{q,s}^j, P_{q,s}^j, t_{q,s}), \quad (37)$$

$$D_{q,s+1,i}^j = \text{Select}(m_{q,s,i}^{\text{sample}}, \bar{D}_{q,s+1,i}^j, D_{q,s,i}^j), \quad (38)$$

$$S_{q,s+1}^j = \text{stopgrad}(P_{q,s}^j E_{\text{tok}}). \quad (39)$$

After at most K fast updates, each unfinished lane privately consolidates what it has learned:

$$\bar{H}_q^j = \text{Pool}\{H_{q,s}^j : 0 \leq s < K\}, \quad (40)$$

$$(\bar{w}_{q+1}^j, \kappa_q^j) = \text{SlowHead}(\Phi_\theta(\text{Pack}(C_b, g_b, r, B^{m,j}, w_{q+1}^j, \bar{H}_q^j, e_m, e_{\text{consolidate}}); \mathcal{A}_q^{m,j})), \quad (41)$$

$$w_{q+1}^j = (1 - \sigma(\kappa_q^j)) \odot w_q^j + \sigma(\kappa_q^j) \odot \bar{w}_{q+1}^j. \quad (42)$$

This slow update remains private to lane j . It does not average lane states or write into the global workspace.

At a window boundary, every unfinished lane forms a refresh bid from public read-only state and only its own consolidated state. Writing $\chi_q^{j+} = (C_b, g, B^{m,j}, w_{q+1}^j, z_{q,K}^j, D_{q,K}^j)$ for lane j 's post-consolidation observation, the dispatcher then solves one joint allocation:

$$\begin{aligned} \{(\mathcal{A}_{q+1}^{m,j}, n_{q+1}^{m,j})\}_{j \in \mathcal{J}_q} = \arg \max_{\{(\mathcal{A}^j, n^j)\} \in \mathcal{F}_{\text{joint}}} \sum_{j \in \mathcal{J}_q} \left[\sum_{v \in \mathcal{A}^j} \ell_v(\chi_q^{j+}) \right. \\ \left. + \sum_{i=1}^{n^j} r_{m,j,i}(\chi_q^{j+}) - \lambda_{\text{switch}} \text{Cost}(\mathcal{A}^j \triangle \mathcal{A}_q^{m,j}) \right], \end{aligned} \quad (43)$$

where $\mathcal{F}_{\text{joint}}$ contains collections of root-connected routes and prefix horizons satisfying equation (4), and $\triangle$ is symmetric difference. The horizon term makes the selected n^j operational rather than a free argmax variable. The implementation uses the projection and estimator of the preceding section. Arbitration receives bids, costs, and route identifiers, not writable lane states; its decisions are returned simultaneously. The switch cost discourages route thrashing but permits a broad route to deepen, fork, or change when private progress reveals a more useful specialist. Refresh never disables the shared substrate or root and never allows one lane to read another lane's private state.

A within-stage route refresh does not reset w^j , z^j , or supported canvas positions, because the brief and private computation remain the same. A new brief, lane, macrocycle, committed block, or abstention performs a full private-state reset. A verification-directed retry may export an identified retry record containing only supported claims, supported token IDs with positions, conflicts, and provenance. The next macrocycle initializes a fresh canvas from that record, draws every disputed position from the declared noise distribution, and clears all self-conditioning. No hidden solver, verifier, latent, or slow state crosses a macrocycle boundary.

4.3 Adaptive private depth

Private halting allocates compute; it cannot authorize publication. For stage m , flatten (q, s) into $n \in \{1, \dots, N_{\text{max}}^{m,j}\}$. A training episode may supervise only some depths. Define the eligible halt set $\mathcal{H}_{m,j}$ to contain depths with a grounded stage-specific output target; every nonterminal member must additionally have a paired next-update target. If $\mathcal{H}_{m,j}$ is empty, that episode contributes no halt loss. With $n^\dagger = \max \mathcal{H}_{m,j}$, the halt logit in equation (39)'s head defines

$$h_n^{m,j} = \begin{cases} \sigma(\eta_n^{m,j}), & n \in \mathcal{H}_{m,j}, n < n^\dagger, \\ 1, & n = n^\dagger, \\ 0, & n \notin \mathcal{H}_{m,j}, \end{cases} \quad (44)$$

$$\rho_n^{m,j} = \prod_{\substack{i \in \mathcal{H}_{m,j} \\ i < n}} (1 - h_i^{m,j}), \quad \pi_n^{m,j} = \rho_n^{m,j} h_n^{m,j}. \quad (45)$$

Thus probability mass cannot disappear into an unlabeled early depth. Solver utility measures the lane’s verified brief contribution; verifier utility measures claim-error discrimination or checker agreement; synthesis utility measures verified integrated-task quality. In every case the stage-conditioned utility $U_{m,j}$ is externally scored. For $n < n^\dagger$, paired continuation gives

$$d_{m,j,n} = U_{m,j}(o_{m,j,n+1}) - U_{m,j}(o_{m,j,n}) - \lambda_C(C_{m,j,n+1} - C_{m,j,n}). \quad (46)$$

Let $\mathcal{H}_{m,j}^- = \mathcal{H}_{m,j} \setminus \{n^\dagger\}$. The eligible halting objective is

$$\begin{aligned} \mathcal{L}_{\text{halt}} = & \sum_{m,j} \frac{1}{|\mathcal{H}_{m,j}| + \epsilon_{\text{num}}} \sum_{n \in \mathcal{H}_{m,j}} \pi_n^{m,j} \left(\ell_{m,j,n}^{\text{out}} + \lambda_{\text{step}} C_{m,j,\leq n} \right) \\ & + \sum_{m,j} \frac{\lambda_{\text{gain}}}{|\mathcal{H}_{m,j}^-| + \epsilon_{\text{num}}} \sum_{n \in \mathcal{H}_{m,j}^-} \rho_n^{m,j} \text{Huber}(\hat{d}_{m,j,n}, d_{m,j,n}), \end{aligned} \quad (47)$$

where empty sums are zero. The compute penalty is applied only at eligible, grounded outputs; sparse labels therefore cannot teach an unpenalized immediate exit.

The next-update bound must remain valid after repeatedly inspecting depths and stopping adaptively. On a disjoint stage-stratified calibration split, run the frozen routing and halting policy and let $\mathcal{I}_e^m$ contain every boundary-valid lane–depth pair that the policy could inspect in calibration episode e . Define one episode-level maximum residual

$$R_e^m = \max_{(j,n) \in \mathcal{I}_e^m} \left(d_{e,m,j,n} - \hat{d}_{e,m,j,n} \right), \quad (48)$$

with empty sets handled by a predeclared omission or pooling rule, and let $q_{1-\alpha_h}^m$ be the finite-sample conformal quantile of the R_e^m values. Then every inspected depth in an exchangeable test episode shares the simultaneous upper confidence bound (UCB)

$$\text{UCB}_{1-\alpha_h}(\hat{d}_{m,j,n}) = \hat{d}_{m,j,n} + q_{1-\alpha_h}^m. \quad (49)$$

The strata, any predeclared low-sample pooling, α_h , and the cumulative-mass threshold are selected before final evaluation; quantiles and the policy are frozen. Episode-level simultaneous coverage and its stage/depth breakdown are both reported. At inference, eligibility is the declared set of boundary-valid readouts rather than label availability. The allocator admits a complete schedule only when $N_{\text{min}}^m \leq N_{\text{cap}}^{m,j} \leq N_{\text{max}}^m$ and its final readout is boundary-valid. Lane (m,j) halts at the first $n \geq N_{\text{min}}^m$ for which cumulative halt mass exceeds the frozen threshold, the boundary is valid, and equation (49) is nonpositive. If that event never occurs, execution ends at $N_{\text{cap}}^{m,j}$ as an explicit *budget cap*, not as a calibrated halt. Halted and budget-capped lanes both break their update loops and become compacted masked no-ops; their termination reasons are logged separately. Categorical sampler stopping, private recursive halting, budget capping, macrocycle continuation, and public commitment are distinct decisions and are evaluated separately.

4.4 Synchronization and global recurrence

Solver lanes execute without cross-lane communication until every lane halts or reaches its cap. A solver barrier materializes identified summaries $\{A^j\}$ simultaneously. Claim-assigned verifier lanes then run the same root-preserving recurrent kernel in parallel under state-isolating masks. A verifier barrier materializes $\{V^k\}$. Only after this second barrier may the global workspace change.

Let $R_r = \text{VerifiedSetEncoder}(\{A^j\}, \{V^k\})$ be permutation-invariant over identified records and mask unsupported branch content. The shared transition performs

$$(\bar{g}_{b,r+1}, \alpha_r) = \text{GlobalHead}\left(\Phi_\theta(\text{Pack}(C_b, g_{b,r}, R_r, e_{\text{global}}); \mathcal{A}_r^{\text{global}})\right), \quad (50)$$

$$g_{b,r+1} = (1 - \sigma(\alpha_r)) \odot g_{b,r} + \sigma(\alpha_r) \odot \bar{g}_{b,r+1}. \quad (51)$$

Thus fast states update at every private microstep, private slow states update at route-window boundaries, and the global workspace updates only after verified fan-in. The new global state may be broadcast into

the next macrocycle, where briefs and routes are redispached. Shared parameters do not imply shared writable state.

The tied transition accumulates gradients from every supervised application. Small experiments use full backpropagation through the declared fast and slow horizons. Scaled experiments declare fast horizon H_f and slow horizon H_s , detach only at those boundaries, and never propagate across a committed public-block boundary. The primary model permits synthesis gradients to pass through structured summaries into their producing lanes within the active horizon. A stop-gradient summary barrier is a required ablation because end-to-end credit may either improve coordination or collapse lane diversity.

4.5 Six functional stages

One root-preserving transition kernel is reused under mode embeddings and information-flow masks. Stages are functions of one parameter system, not six independent transformer stacks.

Framing, decomposition, and the barrier-mediated global update invoke the same Φ_θ with stage-specific projections but no private categorical canvas. Solver, verifier, and synthesis stages use the fully specified private recurrence of Section 4.2; their inputs and utility targets differ, not their parameter system. The functional flow is shown in Figure 3, and the stages are summarized in Table 2.

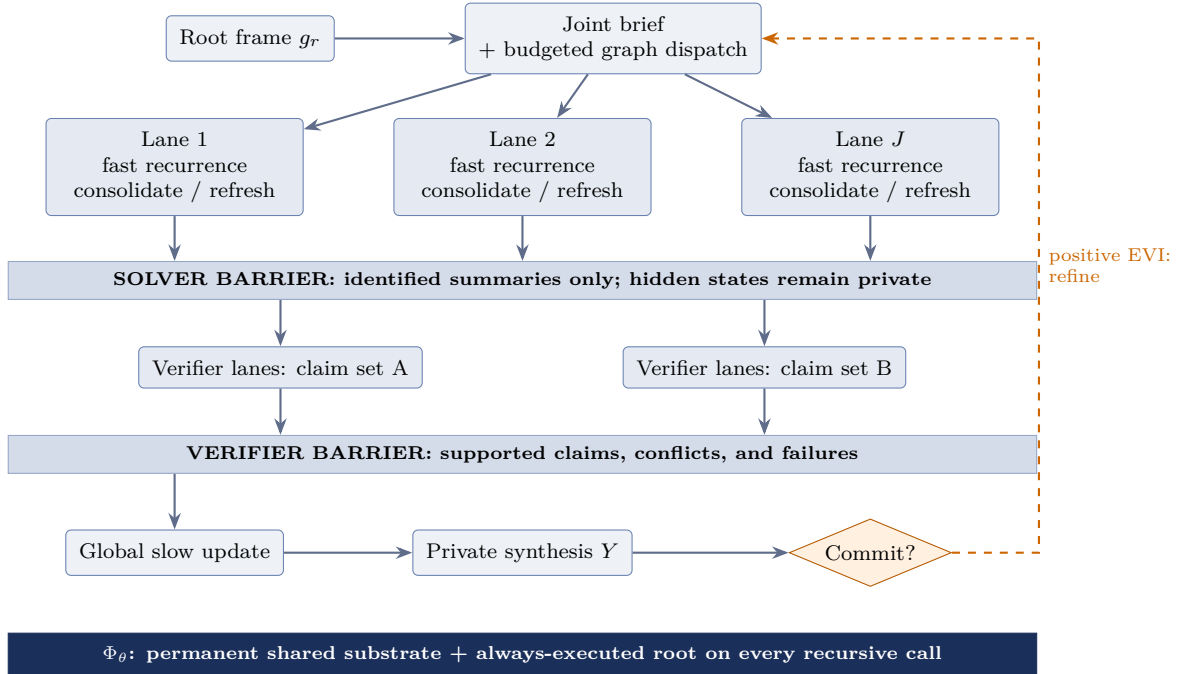


Figure 3: One bounded macrocycle with explicit fan-out, adaptive state-isolated recurrence, barrier-only communication, verified fan-in, and global slow update. All boxes reuse one parameter system. Only the separately synthesized and scored token snapshot may become public.

The architecture can collapse to a cheap path by selecting one lane, no specialists, and few refinement steps, but commitment remains mandatory.

4.6 End-to-end native execution

The following is tensor-program pseudocode for one model, not a collection of processes. “Committed” denotes the only externally visible state; “private read-only” denotes barrier-approved internal records that cannot be mutated by a lane. A reservation is a draw-down subledger: every `ExecuteReserved` call fails before execution if the remaining balance is insufficient, then atomically settles measured cost, and settled units cannot be reused. `ChargeControl` applies the same rule to routing, allocation, assignment, record construction, and decisions; constant-time ledger bookkeeping is not a learned model operation. `RunPrivateRecurrence` applies the rule internally to every fast and consolidation call. `PartitionReserved` divides one prior reservation into disjoint named balances; it does not reserve the

Table 2: Macrocycle stages and their variable specialist use. The shared trunk and root execute in every row.

Stage	Function	Conditional computation
Framing	Extract objective, constraints, uncertainty, and response contract into g .	Root-preserving recurrence; usually a shallow or root-only route.
Decomposition	Select active lanes and create distinct briefs.	Joint dispatch under a global lane, expert, and step budget.
Parallel solving	Refine state-isolated fast states, canvases, and private slow states.	Per-lane route windows, adaptive depth, and lane-conditioned refresh under joint budget arbitration.
Verification	Test atomic claims, constraints, evidence, code, or counterexamples.	Parallel state-isolated verifier recurrence with claim-conditioned routes.
Synthesis	Integrate identified summaries and verified findings into private Y .	Barrier-mediated fan-in followed by a finding-conditioned route.
Commitment	Estimate explicit risks and continuation value for the exact Y snapshot.	Root plus calibration-targeted risk and EVI heads; optional checkers remain budgeted.

same budget again. The control pool is claimed once when a cycle opens; thereafter `downstream_min` is the sum of the remaining summary, verification, global-update, synthesis, commitment-transition, scoring, and possible-publication minima.

Three forms of concurrency are distinct. Within one active expert subgraph, nodes at the same graph depth may execute together, while ancestor–descendant edges impose only the selected graph’s critical path. Across solver or verifier lanes, all unfinished lanes at a microstep are folded into the batch dimension and advance against the same frozen workspace without reading one another’s writable state. Across token positions, categorical denoising revises a private block bidirectionally. None of these forms allows streaming public output: barriers expose only identified records, and publication remains a single exact-snapshot transition.

The execution has three corresponding depths. Graph depth selects how far specialization descends; private recurrent depth determines how long each lane continues; macrocycle depth determines whether verified findings justify another complete round. The joint ledger couples all three, while independent halt decisions let cheap lanes stop without forcing difficult lanes to stop. The root-preserving graph operator is applied on every charged recursive call, so deeper specialists supplement a continuously active general substrate rather than replacing it.

Listing 1: Budgeted dispatch and state-isolated parallel solving.

```

C and K_public remain immutable during private computation
ledger ← BeginBlockLedger(B_block)
fallback_escrow ← ReserveOnce(
  ledger, fallback_generate_min + fallback_commit_min
    + fallback_score_min + fallback_publish_min
    + fallback_control_min)
commit_min ← commit_transition_min + score_min + publish_min

frame_control ← Reserve(ledger, frame_control_min)
frame_bid ← ChargeControl(frame_control, RouteBid(
  mode=frame, committed=C, private_read_only=previous_g))
frame_route, frame_reservation ← ChargeControl(
  frame_control, AllocateOne(
    frame_bid, ledger, reserve=macrocycle_min))
g ← ExecuteReserved(frame_reservation, SharedRecur(
  mode=frame, committed=C, private_read_only=previous_g,
  experts=root + frame_route))

for macrocycle r in 1..R_max while ledger permits:
  cycle ← OpenSubledger(
    ledger, minimum=control_min + decompose_min + solve_min + summary_min
      + verify_min + global_min + synth_min + commit_min)
  control ← ClaimNamedReserve(cycle, control_min)
  dec_bid ← ChargeControl(control, RouteBid(
    mode=decompose, committed=C,
    private_read_only=frozen(g)))
  dec_route, dec_reservation ← ChargeControl(

```

```

    control, AllocateOne(dec_bid, cycle))
briefs, lane_bids ← ExecuteReserved(dec_reservation, SharedRecur(
    mode=decompose, committed=C, private_read_only=frozen(g),
    experts=root + dec_route))
routes, window_plans, active_lanes ← ChargeControl(
    control, JointAllocateSchedules(
        lane_bids, cycle, reserve=downstream_min))

parallel for solver lane j in active_lanes:
    w_j, z_j, D_j, S_j ← ExecuteReserved(
        window_plan_j.initialize, Initialize(
            mode=solve, C, frozen(g), brief_j, lane_key_j))

for route window q in 1..Q_max:
    parallel for unfinished solver lane j:
        for scheduled microstep s in window_plan_j.fast_steps:
            H_j ← ExecuteReserved(window_plan_j.kernel_s,
                SharedTransition(
                    mode=solve, committed=C,
                    private_read_only=frozen(g),
                    private_lane=(brief_j, w_j, z_j, D_j, S_j),
                    experts=root + route_j))
            z_j, D_j, S_j, halt_j, gain_j ←
                ExecuteReserved(
                    window_plan_j.fast_head_s, FastUpdate(H_j))
            if PrivateHalt(halt_j, gain_j, D_j):
                Freeze(lane_j); break
        if lane j is unfinished:
            w_j ← ExecuteReserved(window_plan_j.consolidate,
                PrivateConsolidate(own_state_j, root + route_j))
            refresh_bid_j ← ChargeControl(
                control, BidFromOwnState(own_state_j, route_j))

CompactHaltedLanes()
routes, next_window_plans ← ChargeControl(
    control, JointRefresh(
        all refresh_bids, cycle, reserve=downstream_min))
parallel for unfinished lane j without next_window_plan_j:
    Freeze(lane_j, reason=BUDGET_CAP)
window_plans ← next_window_plans
if every solver lane is frozen: break

parallel for unfinished solver lane j:
    Freeze(lane_j, reason=WINDOW_CAP)

SOLVER_BARRIER
summary_reservations ← ReserveParallel(cycle, active_lanes)
parallel for solver lane j:
    A_j ← ExecuteReserved(summary_reservation_j,
        IdentifiedSummary(own_state_j, route_history_j))

```

Listing 2: Barrier-mediated verification, synthesis, and atomic commitment.

```

assignments ← ChargeControl(
    control, AssignAtomicClaims(A, briefs, frozen(g)))
verify_bids ← ChargeControl(
    control, RouteBids(mode=verify, assignments, A, frozen(g)))
verify_routes, verify_reservations, verifier_lanes ←
    ChargeControl(control, JointAllocate(
        verify_bids, cycle,
        reserve=global_min + synth_min + commit_min))
parallel for verifier lane k in verifier_lanes:
    V_k ← RunPrivateRecurrence(
        mode=verify, committed=C,
        private_read_only=(frozen(g), A, assignments_k),
        private_lane=Initialize(mode=verify, assignment_k),
        experts=root + verify_route_k,
        reservation=verify_reservation_k)

VERIFIER_BARRIER

```



```

verified ← ChargeControl(control, VerifiedSet(A, V))
global_bid ← ChargeControl(control, RouteBid(
  mode=global, committed=C,
  private_read_only=(frozen(g), verified)))
global_route, global_reservation ← ChargeControl(
  control, AllocateOne(
    global_bid, cycle, reserve=synth_min + commit_min))
g_candidate ← ExecuteReserved(global_reservation, GlobalSlowUpdate(
  committed=C, private_read_only=(frozen(g), verified),
  experts=root + global_route))

contract ← ChargeControl(
  control, IntegrationContract(verified, conflicts))
synth_bid ← ChargeControl(control, RouteBid(
  mode=synthesize, committed=C,
  private_read_only=(g_candidate, A, V, contract)))
synth_route, synth_reservation ← ChargeControl(
  control, AllocateOne(
    synth_bid, cycle, reserve=commit_min))
Y ← RunPrivateRecurrence(
  mode=synthesize, committed=C,
  private_read_only=(g_candidate, A, V, contract),
  private_lane=Initialize(mode=synthesize, contract),
  experts=root + synth_route, reservation=synth_reservation)

precommit ← ChargeControl(control, FreezeCandidateInputs(
  context_version, C, g_candidate, A, V, contract,
  token_ids(Y), valid_length(Y), SynthesisReadout(Y),
  route_history, calibration_version))
commit_kernel_reservation, score_reservation,
publish_reservation ← PartitionReserved(
  ClaimNamedReserve(cycle, commit_min),
  commit_transition_min, score_min, publish_min)
commit_bid ← ChargeControl(control, RouteBid(
  mode=commit, committed=C, private_read_only=precommit))
commit_route ← ChargeControl(control, RouteWithinReserved(
  commit_bid, commit_kernel_reservation))
s_commit ← ExecuteReserved(commit_kernel_reservation,
  SharedTransition(mode=commit, committed=C,
    private_read_only=precommit,
    experts=root + commit_route))
snapshot ← ChargeControl(
  control, Freeze(precommit, s_commit, commit_route))
risks, EVI ← ExecuteReserved(score_reservation,
  ScoreExactSnapshot(snapshot))
decision ← ChargeControl(control,
  DecideExactSnapshot(risks, EVI, snapshot, ledger))
if decision == COMMIT:
  ExecuteReserved(publish_reservation,
    AtomicPublishAndCausalExtend(snapshot, C, K_public))
  DestroyAllPrivateState(); return
if decision == REFINES:
  retry_record ← ChargeControl(control,
    IdentifiedRetryRecord(snapshot, verified, conflicts))
  DestroyAllPrivateState()
  g ← ChargeControl(
    control, RetryFrame(C, g_candidate, retry_record))
  continue
DestroyAllPrivateState(); break

```

Listing 3: Single-escrow causal fallback and guarded publication.

```

fallback_generate_reservation, fallback_commit_reservation,
  fallback_score_reservation, fallback_publish_reservation,
  fallback_control_reservation ←
    PartitionReserved(ActivateEscrow(fallback_escrow),
      fallback_generate_min, fallback_commit_min,
      fallback_score_min, fallback_publish_min,
      fallback_control_min)
fallback_Y ← ExecuteReserved(fallback_generate_reservation,

```

```

RunPrivateCausalFallback(C)
fallback_precommit ← ChargeControl(
    fallback_control_reservation, FreezeCandidateInputs(
        context_version, C, token_ids(fallback_Y),
        valid_length(fallback_Y), calibration_version))
fallback_s_commit ← ExecuteReserved(fallback_commit_reservation,
    SharedTransition(mode=commit, committed=C,
        private_read_only=fallback_precommit, experts=root))
fallback_snapshot ← ChargeControl(fallback_control_reservation,
    Freeze(fallback_precommit, fallback_s_commit, route=root))
fallback_risks ← ExecuteReserved(fallback_score_reservation,
    ScoreFallbackExactSnapshot(fallback_snapshot))
if ChargeControl(fallback_control_reservation,
    AcceptFallback(fallback_risks, fallback_snapshot)):
    ExecuteReserved(fallback_publish_reservation,
        AtomicPublishAndCausalExtend(
            fallback_snapshot, C, K_public))
    return
abstain

```

5. Diffusion as the Private Execution Mechanism

5.1 Shared causal encoder and bidirectional private decoder

A candidate implementation tests a same-weight dual-mode pattern. Committed context is encoded causally and stored in a read-only key-value cache. Private solver and synthesis canvases use bidirectional attention within each canvas while attending to the committed cache. Compatible encoder and decoder layers share parameters; stage, lane, and route embeddings distinguish their functions. Weight sharing is intended to reuse pretrained representations and avoid duplicate parameter sets, but preservation of capability across modes is an empirical hypothesis. DiffusionGemma demonstrates model-scale feasibility for a causal encoder, a bidirectional diffusion decoder, self-conditioning, and sparse experts; it does not prove that the same choice is optimal for HLWM [4]. Shared weights must be compared with mode-specific weights or adapters, including causal-fallback quality after diffusion training.

5.2 Categorical corruption and refinement

Text is discrete, so private block corruption is categorical. Let ν be a declared vocabulary-noise distribution and $\beta_t \in (0, 1)$. A simple multinomial forward process is

$$Q_t = (1 - \beta_t)I + \beta_t \mathbf{1}\nu^\top, \quad (52)$$

$$\bar{Q}_t = Q_1 Q_2 \cdots Q_t, \quad (53)$$

$$q(x_t | x_0) = \text{Cat}(\text{onehot}(x_0) \bar{Q}_t). \quad (54)$$

The denoiser predicts clean tokens conditioned on the committed cache, global frame, lane brief, private latent state, active expert subgraph, and the previous clean-token distribution:

$$p_\theta(x_0 | x_t, t, C, g, B^j, z^j, \mathcal{A}^{m,j}, S_{t+1}). \quad (55)$$

The analytic categorical posterior combines this clean prediction with the forward process to sample x_{t-1} . Self-conditioning embeds the previous clean-token distribution rather than a sampled string. The mechanism is grounded in discrete diffusion and self-conditioning work [5, 6, 7, 8]; the exact schedule and sampler remain implementation hyperparameters.

Categorical sampler stopping is a token-update mask, not a reasoning or publication decision. For position i , let

$$H_{q,s,i}^j = - \sum_x P_{q,s,i}^j(x) \log P_{q,s,i}^j(x), \quad \delta_{q,s,i}^j = \|P_{q,s,i}^j - P_{q,s-1,i}^j\|_1. \quad (56)$$

With development-selected thresholds τ_H, τ_δ and patience p_{sample} , let $c_{q,s,i}^j = 1$ only after both $H_{q,s,i}^j \leq \tau_H$ and $\delta_{q,s,i}^j \leq \tau_\delta$ hold for p_{sample} consecutive eligible sampler steps. Let $r_{q,s,i}^{\text{renoise}}$ mark an explicit re-noise

command, and let $m_{\text{prev},i}^{\text{sample}}$ be the mask at the preceding executed step, carried across route windows and initialized to one. The recursive rule is

$$m_{q,s,i}^{\text{sample}} = r_{q,s,i}^{\text{renoise}} \vee \left(m_{\text{prev},i}^{\text{sample}} \wedge \neg c_{q,s,i}^j \right). \quad (57)$$

Thus zero is absorbing unless an explicit re-noise command reactivates the position and resets its patience counter. Equation (39) leaves stopped positions unchanged while other positions continue to denoise. When every position is stopped, the categorical canvas no longer changes, but latent recurrence, consolidation, verification, and the private halt rule may still run. Re-noising assigns a new timestep only to identified disputed positions. Thresholds are frozen before test evaluation, and reports include saved sampler steps and clean-token error; low entropy or stability is never interpreted as semantic correctness.

5.3 Sampler stopping is not semantic commitment

Entropy-bounded or stability-based stopping can save denoising steps [4, 10], but a stable canvas may be confidently wrong. Entropy therefore governs only whether another sampler step is likely to change the private canvas. It is not evidence of semantic correctness.

Commitment is itself a root-preserving model invocation. Let $\mathcal{I}_r^{\text{pre}}$ be an immutable candidate-input record containing committed context, the global candidate state, identified summaries, verification records, integration contract, synthesis readout, exact candidate token IDs, route provenance, and calibration versions. The commitment state is

$$s_r = \text{CommitStateHead} \left(\Phi_{\theta} \left(\text{Pack}(\mathcal{I}_r^{\text{pre}}, e_{\text{commit}}); \mathcal{A}_r^{\text{commit}} \right) \right), \quad v_0 \in \mathcal{A}_r^{\text{commit}}. \quad (58)$$

Thus the shared substrate and root execute before the risk and continuation heads; any conditional commitment specialist remains subject to the same connected-route budget. For task x , let $\mathcal{K}(x)$ be a preregistered set of observable failure predicates, such as hard-constraint violation, an unsupported assertion, a failed executable check, or omission of a required deliverable. For frozen synthesis block Y_r and commitment state s_r , the risk heads estimate

$$q_k(s_r, Y_r) = \Pr(E_k(Y_r, x) = 1 \mid s_r, Y_r), \quad k \in \mathcal{K}(x). \quad (59)$$

Each head is trained with a proper scoring loss and calibrated on a disjoint held-out split: $\bar{q}_k = \text{Cal}_k(q_k)$. A threshold τ_k is selected so that a one-sided confidence bound on empirical failure among emitted examples does not exceed preregistered risk limit α_k at the target coverage. This selection uses final emissions produced by the frozen complete decision policy, not a pool of intermediate candidates, so adaptive snapshot selection is represented in calibration. Calibration maps and thresholds are frozen before test evaluation; the paper targets calibration rather than assuming it.

Semantic stopping additionally predicts the value of one more permitted macrocycle. For task utility U and measured compute C , paired rollouts provide target

$$d_r = U(Y_{r+1}, x) - U(Y_r, x) - \lambda_C(C_{r+1} - C_r), \quad \widehat{\text{EVI}}_r = \mathbb{E}[d_r \mid s_r, Y_r]. \quad (60)$$

A disjoint macrocycle-calibration split makes the continuation bound operational under repeated adaptive checks. Run the frozen complete policy and let $\mathcal{I}_e^E$ contain every macrocycle at which episode e could inspect continuation value. Define the episode-level maximum residual

$$R_e^E = \max_{r \in \mathcal{I}_e^E} (d_{e,r} - \widehat{\text{EVI}}_{e,r}), \quad (61)$$

with empty sets handled by a predeclared rule, and let $q_{1-\alpha_E}^E$ be its frozen finite-sample conformal quantile. Then

$$\text{UCB}_{1-\alpha_E}(\widehat{\text{EVI}}_r) = \widehat{\text{EVI}}_r + q_{1-\alpha_E}^E. \quad (62)$$

The calibration split, utility, compute price, α_E , complete decision policy, and any predeclared task-family pooling are frozen before test evaluation. The maximum-residual score gives simultaneous episode-level coverage over the inspected macrocycles under exchangeability, so stopping at the first nonpositive bound does not silently convert a marginal guarantee into an invalid adaptive one. Simultaneous coverage and per-macrocycle diagnostics are reported.

The model commits only when

$$\text{Commit}(Y_r) = \mathbf{1}[\text{Boundary}(Y_r)] \prod_{k \in \mathcal{K}(x)} \mathbf{1}[\bar{q}_k \leq \tau_k] \mathbf{1}[\text{UCB}_{1-\alpha_E}(\widehat{\text{EVI}}_r) \leq 0]. \quad (63)$$

If commitment fails, another macrocycle is permitted only when

$$\text{Refine}(Y_r) = \mathbf{1}[\text{UCB}_{1-\alpha_E}(\widehat{\text{EVI}}_r) > 0] \mathbf{1}[B_{\text{unreserved}} \geq B_{\text{next,min}}]. \quad (64)$$

Thus a failed risk check does not automatically trigger more computation: positive calibrated continuation value and a complete next-cycle minimum are both required. If neither equation (63) nor equation (64) holds, the model stops private diffusion and either uses its separately escrowed causal fallback or abstains. Targeted re-noising and the fallback are hypotheses to compare with full-canvas restart and a dedicated causal control, not assumed improvements.

5.4 Atomic publication and causal cache reconstruction

The decoder first freezes $\mathcal{I}_r^{\text{pre}}$, executes the budgeted commitment transition, and then, before either scoring head is evaluated, freezes an immutable snapshot $\Sigma_b = (v_b, y_b^*, L_b, \mathcal{I}_b)$ containing the committed-context version, exact decoded token IDs, valid length, and every representation consumed by risk or expected-value-of-information scoring. In particular, $\mathcal{I}_b$ stores immutable copies (plus content hashes) of $\mathcal{I}_r^{\text{pre}}$, s_r , the commitment route, and calibration-version identifiers. A hash without the underlying scoring inputs is insufficient. Commitment authorizes only Σ_b ; any changed input creates a new snapshot and invalidates the decision. The exact buffered IDs y_b^* are published without resampling or re-decoding.

Let K_b be the causal key-value cache for committed context C_b . Publication is the atomic transition

$$(C_b, K_b) \longrightarrow (C_b \parallel y_b^*, \text{CausalExtend}_{\theta}(K_b, y_b^*)), \quad (65)$$

conditioned on the live context version still equaling v_b . Cache extension is computed privately and becomes visible together with the new public context; on failure, neither changes. Private bidirectional keys and values, stage embeddings, lane states, summaries, and verifier representations are never inserted into K_b . The accepted IDs are run through the same model under a strict causal mask before generation continues. If prefix-measurable cache extension cannot be guaranteed, the implementation must causally re-encode the complete committed context. Rejection leaves (C_b, K_b) unchanged.

In plain terms: the model works in pencil in private, and the only way ink reaches the page is a frozen, checksummed snapshot that has passed every calibrated risk check, or else an explicitly safe fallback sentence.

6. Learning the Architecture

6.1 Training data must supervise outcomes, not fictional thoughts

Foundation data and workspace data serve different purposes. Foundation corpora teach language, knowledge, mathematics, and code. Workspace episodes teach decomposition, complementary work, verification, integration, and calibration-targeted stopping. Table 3 gives the required observable structure.

For coding, a strong episode contains repository context, jointly generated scopes, state-isolated patches or analyses, compiler and test output, verifier findings, an integrated patch, and final test results. Route decisions, specialist strengths, lane count, and step budgets should be learned from outcomes and compute pressure. Human-readable expert annotations may be weak priors, but invented confidence scores and handwritten dominance schedules are not ground truth.

Programmatically checkable records carry a structured answer specification rather than an exact target phrase. The current narrow graders compare the final numeric value for arithmetic, both value and unit for conversions, the final numeric sequence for ordering, and an explicit missing-evidence predicate with no unsupported numeric guess for abstention. Thus “The answer is 36360” and another wording with the same checked value receive the same outcome label. These graders authorize only their declared task class; they do not turn lexical similarity into a correctness judgment for open-domain answers.

Table 3: Training record families. Synthetic records are useful only when their outcomes are mechanically or independently checked.

Record family	Contents	Grounding signal
Foundation	Documents, mathematics, code, documentation, and dialogue.	Source quality, deduplication, licensing, contamination controls.
Workspace	Problem, constraints, candidate briefs, complementary or competing work, and dependencies.	Coverage, non-redundancy, task outcome, expert intervention tests.
Verification	Claims, counterexamples, failed constraints, compiler output, tests, proof checks, or cited evidence.	Executable or independently adjudicated result.
Synthesis and commitment	Integrated answer, rejected alternatives, corrections, and whether another macro-cycle changed or improved the result.	Final task score, constraint satisfaction, regression tests, failure-event labels, and paired continuation utility.

A supported abstention is a candidate public answer, not the absence of an answer. When required evidence is missing, the clean target is a concise statement that the value cannot be determined from the supplied information, and a guessed value is its paired negative. If that clean target passes the exact-snapshot policy, it is atomically published like any other answer. If a generated candidate fails the policy, the candidate remains private and the public interface returns a fixed safe fallback. This separates *semantic abstention* from *policy withholding* and avoids teaching the commitment head that honest uncertainty should produce no user-visible response.

6.2 Training objective and eligible supervision

Let $\mathcal{D}_F$ contain foundation text, $\mathcal{D}_W$ workspace episodes with accepted briefs and externally scored lane outcomes, $\mathcal{D}_V$ atomic claims with adjudicated verification labels, $\mathcal{D}_S$ verified synthesis targets, and $\mathcal{D}_P$ paired macrocycle rollouts. A record contributes to an auxiliary loss only when its required target is present; missing targets are not replaced by invented latent traces.

The causal and diffusion objectives are

$$\mathcal{L}_{\text{causal}} = -\mathbb{E}_{(x,y) \sim \mathcal{D}_F} \sum_i \log p_{\theta}(y_i \mid y_{<i}, x), \quad (66)$$

$$\mathcal{L}_{\text{denoise}} = -\mathbb{E}_{(x_0,c), t, x_t} \sum_i \log p_{\theta}(x_{0,i} \mid x_t, c, t), \quad (67)$$

where clean blocks for the second expectation come from foundation, workspace, or verified synthesis records, t follows the declared time distribution, and $x_t \sim q(x_t \mid x_0)$. Sampling one time and corruption realization per example gives the Monte Carlo estimator.

6.3 Hybrid one-pass training for the tied private transition

DiffusionBlocks interprets network depth as a denoising trajectory and trains depth/noise regions without end-to-end backpropagation through the complete chain [9]. Its recurrent-language experiment is the closest precedent for the repeatedly reused transition in HLWM, but fixed independently parameterized noise-range blocks would violate the shared-transition invariant and would not train routing, barriers, synthesis, or commitment jointly. We therefore use the insight only for the fast private recurrence.

Let $\pi(t)$ be a declared categorical-time distribution; the prototype uses probability proportional to the incremental corruption mass $\bar{\alpha}_{t-1} - \bar{\alpha}_t$. For one lane target x_0^j , one local update samples $t \sim \pi$, corrupts $x_t^j \sim Q_t(\cdot \mid x_0^j)$, and evaluates

$$\hat{\mathcal{L}}_{\text{local}} = -\frac{1}{\sum_i m_i^j} \sum_i m_i^j \log p_{\theta}(x_{0,i}^j \mid x_t^j, C, B^j, t). \quad (68)$$

This x_0 -prediction update executes the tied transition once and does not construct the unused reverse posterior. It supplies local gradients to timestep conditioning, private fast state, briefs, root and routed adapters, and the denoising head. It does *not* supply the cross-step or cross-stage credit needed by private

consolidation, route refresh, the summary barrier, global recurrence, synthesis, verification, halting, risk, or publication.

Training is consequently hybrid: many single-pass local updates are interleaved with causal-language updates and short explicit joint unrolls. The required controls are full backpropagation through the reverse horizon, declared truncated backpropagation, single-pass local training, and single-pass pretraining followed by the same joint tuning budget. Increasing the nominal number of blocks is not assumed to help; each learned subproblem must retain sufficient depth and the comparison must match tokens, updates, and inference compute. No evidence in the present prototype supports splitting the six functional stages or the expert graph into independently trained blocks.

On memory, the local phase removes storage of activations for K recurrent applications, changing that component from approximately $O(KA)$ to $O(A)$ for activation size A . The shared Qwen weights, root, conditional modules, gradients, and optimizer state remain resident, and joint unrolls still retain their declared horizon. This is an activation-memory and backpropagation-compute saving for one transition, not a whole-model $K \times$ memory reduction or a communication-free training claim.

For records containing an accepted structured brief B^* ,

$$\mathcal{L}_{\text{brief}} = -\mathbb{E}_{\mathcal{D}_W} \log p_{\theta}(B^* | C, g). \quad (69)$$

Branch work without a verified demonstration is trained from external outcome reward R_j using a score-function estimator,

$$\mathcal{L}_{\text{work}} = -\mathbb{E}_{\mathcal{D}_W} \sum_j (R_j - b(x)) \log p_{\theta}(W_j | B^j, C, g), \quad (70)$$

with learned baseline b . Demonstration likelihood may be added only when a verified lane-level target W_j^* exists.

Writing BCE for binary cross-entropy, for adjudicated claim label $e_c \in \{0, 1\}$ and verified integrated target Y^* ,

$$\mathcal{L}_{\text{verify}} = \mathbb{E}_{(c, e_c) \sim \mathcal{D}_V} \text{BCE}(q_{\theta}(c), e_c), \quad (71)$$

$$\mathcal{L}_{\text{synth}} = -\mathbb{E}_{(A, V, Y^*) \sim \mathcal{D}_S} \log p_{\theta}(Y^* | A, V, C, g). \quad (72)$$

Commitment uses proper scoring for each available task-specific failure event and regression to the observed compute-adjusted continuation value:

$$\mathcal{L}_{\text{commit}} = \mathbb{E}_{\mathcal{D}_P} \left[\sum_{k \in \mathcal{K}(x)} \text{BCE}(q_k, e_k) + \lambda_E \text{Huber}(\widehat{\text{EVI}}, d_r) \right]. \quad (73)$$

Probability calibration is fitted afterward on a disjoint calibration split; low training loss is not called calibration.

Graph connectedness and the per-example hard budget are enforced by the execution-cost projection. The remaining routing regularizer is

$$\mathcal{L}_{\text{route}} = \lambda_{\text{cost}} \mathbb{E} \left[\frac{\widehat{C}_{\theta}(x)}{B(x)} \right] + \lambda_{\text{tail}} \left[\text{CVaR}_{0.95}(\widehat{C}_{\theta}) - B_{0.95} \right]_+ + \lambda_{\text{cap}} \sum_v [n_v - \text{cap}_v]_+^2, \quad (74)$$

where $\widehat{C}_{\theta}$ uses the relaxed gates only as a training estimator and n_v is the routed batch load. Capacity penalties prevent overload but do not force uniform expert use. Downstream gradients reach gates through the straight-through estimator; a score-function router is evaluated as a control because the straight-through gradient is biased.

Every permitted halt depth must expose an evaluable state, but its target must match the lane's role. Let $\ell_{r,n}^j$ be a brief-specific loss at private update n , ℓ_r^g a global or synthesis loss after macrocycle r , and masks $m_{r,n}^j, m_r^g \in \{0, 1\}$ indicate that the required externally grounded target exists. The primary estimator sets $w_{r,n}^j = w_r^g = 1$ for every observed eligible checkpoint. If checkpoints are subsampled, w is the clipped inverse inclusion probability fixed by the sampling design; it is stop-gradient and never a function of the realized loss or learned halt probability. Role-aligned intermediate supervision is

$$\mathcal{L}_{\text{deep}} = \mathbb{E} \left[\frac{\sum_{r,j,n} m_{r,n}^j w_{r,n}^j \ell_{r,n}^j + \sum_r m_r^g w_r^g \ell_r^g}{\sum_{r,j,n} m_{r,n}^j w_{r,n}^j + \sum_r m_r^g w_r^g + \epsilon_{\text{num}}} \right]. \quad (75)$$

Solver checkpoints use only verified brief-specific demonstrations, executable outcomes, or adjudicated claim labels; they are not forced to reproduce the same final answer. Verifier checkpoints use proper scoring against adjudicated error labels, while global and synthesis checkpoints may use a verified integrated target. Normalization prevents a long-running example from receiving more gradient merely because it executed more steps. Final-only, common-final-target-at-every-depth, and detached-survival-weighted supervision are required controls because aggressive early supervision may cause premature convergence or homogenize the lanes.

The complete base objective is

$$\mathcal{L}_{\text{base}} = \sum_{a \in \{\text{causal}, \text{denoise}, \text{brief}, \text{work}, \text{verify}, \text{synth}, \text{commit}, \text{route}, \text{halt}, \text{deep}\}} \lambda_a \mathcal{L}_a. \quad (76)$$

Loss weights, data-mixture proportions, risk limits, and compute prices are selected on development data and frozen before final evaluation. Let $\bar{\Delta}_k^{\text{root}} = \max_{q \in \mathcal{Q}_{\text{halt}}} \Delta_{k,q}^{\text{root}}$ and $\bar{\Delta}_k^{\text{route}} = \max_{q \in \mathcal{Q}_{\text{halt}}} \Delta_{k,q}^{\text{route}}$, with the terminal readout used when a stage has no adaptive exit. The behavioral constraints of Section 3.2 are incorporated with nonnegative dual variables:

$$\begin{aligned} \mathcal{J}(\theta, \mu) = \mathcal{L}_{\text{base}} &+ \sum_k \mu_k^{\text{root}} \left[\bar{\Delta}_k^{\text{root}} - \epsilon_k^{\text{root}} \right]_+ \\ &+ \sum_k \mu_k^{\text{route}} \left[\bar{\Delta}_k^{\text{route}} - \epsilon_k^{\text{route}} \right]_+. \end{aligned} \quad (77)$$

The multipliers are updated by dual ascent or another constrained-optimization method. Continued foundation training and optional distillation help optimize the anchor objectives, but neither is treated as proof that general capabilities were retained. Losses are normalized by eligible checkpoints, the global gradient norm is clipped, and activation checkpointing changes memory use but not the objective. Full-unroll and declared truncated-horizon results, including gradient-norm distributions, must be reported.

6.4 Prototype numerical regime and narrow policy calibration

The T4 prototype keeps the transplanted Qwen tensors frozen in the resolved autocast dtype (FP16, or BF16 where the venue supports it; Experiment 4 recorded a preregistration deviation when `auto` resolved to BF16 on Turing) and trains separate FP32 low-rank sidecars plus new HLWM modules. It computes the full 151,936-way token softmax and cross-entropy in FP32, excludes padded rows before vocabulary loss, uses boolean attention validity masks, and forces padded hidden states back to zero. A real-checkpoint preflight executes the exact context, canvas, brief, and vocabulary dimensions with named forward hooks, a backward pass, and finite-gradient assertions before either training seed launches. These are numerical correctness conditions, not architectural contributions.

After training, the narrow commitment prototype freezes the model and scores clean and mechanically corrupted candidates from the verified validation anchors. A deterministic threshold search fits the conjunction of minimum commitment probability, maximum risk probability, and maximum verifier-error probability, subject to a clean-acceptance floor. The test split is not consulted, the fitted thresholds are embedded in both checkpoint and adapter metadata, and held-out clean acceptance and corrupt rejection are reported separately. Because these anchors cover only arithmetic, unit conversion, ordering, and missing evidence, passing this calibration gate does not imply calibrated risk on open-domain generations.

6.5 Progressive training

Training every mechanism simultaneously from random initialization creates an ill-conditioned credit-assignment problem. A practical curriculum is:

1. pretrain the shared causal/block-diffusion substrate and root on foundation data;
2. tie the root-preserving transition across a short fixed recursive horizon and supervise every eligible readout;
3. introduce the expert graph, private slow states, and route-window refresh under budget and anchor constraints;

4. train jointly briefed, state-isolated parallel lanes before enabling adaptive private halting;
5. add routed parallel verification, barrier-mediated global updates, and cross-lane synthesis using observable outcomes;
6. train task-specific failure risks, macrocycle EVI, and the full variable-depth computation jointly;
7. jointly tune the full macrocycle, followed by outcome- or preference-based post-training.

This schedule does not make the stages separate deployed models; it is a stabilization strategy for one final parameter system.

7. Computation, Memory, and Scaling

Total stored parameters, unique parameters touched, parameter executions, and FLOPs are different quantities. Dormant experts save computation, not automatically storage. A fully resident deployment must hold the complete graph; an offloaded deployment trades memory for expert-loading latency. Let $\mathcal{Q}(x)$ be the set of executed stage-lane-refinement calls for input x , and let Θ_q be the exact set of parameter scalars read by call q . Unique parameter touch is

$$P_{\text{touch}}(x) = \left| \bigcup_{q \in \mathcal{Q}(x)} \Theta_q \right|. \quad (78)$$

When shared modules, heads, and expert banks are parameter-disjoint, this becomes

$$P_{\text{touch}}(x) = P_{\text{shared}} + P_{\text{heads}}(x) + \sum_{v \in \bigcup_{q \in \mathcal{Q}(x)} \mathcal{A}_q(x)} P_v. \quad (79)$$

Training memory must be reported with the same separation. For unroll horizon H , a useful accounting identity is

$$M_{\text{train}}(H) = M_{\text{shared}/\text{root}} + M_{\text{conditional}} + M_{\text{grad}} + M_{\text{optimizer}} + M_{\text{act}}(H) + M_{\text{communication}}. \quad (80)$$

Single-pass local training reduces only $M_{\text{act}}(H)$ and the associated backward graph by setting $H = 1$ for the fast recurrence. It does not remove resident shared or expert weights, gradients, optimizer state, or inference-time model loading. Reports distinguish allocated from reserved device memory and state whether two GPUs perform data/model parallelism or independent replications. The reported Experiment 1–4 dual-T4 experiments used one complete independent seed per GPU; they were not two-GPU distributed executions of one model. The preregistered Qwen3-8B experiment instead uses two-process distributed data parallelism, with one quantized replica and one local batch on each T4, to optimize one shared candidate checkpoint.

An expert used by several lanes or refinement steps is counted once in P_{touch} , but every invocation contributes FLOPs:

$$F_{\text{exec}}(x) = F_{\text{encode}}(C) + \sum_{q \in \mathcal{Q}(x)} F_q + \mathbf{1}[\text{Commit}(Y)] F_{\text{causal-encode}}(Y). \quad (81)$$

The final term accounts for rebuilding causal key-value state from committed token IDs. Batching lanes may reduce latency but does not reduce executed FLOPs. Reports therefore separate stored parameters, unique parameter touch, FLOPs, mean and 95th-percentile (P95) latency, peak memory, and communication volume.

Lane states are folded into the batch dimension so every unfinished solver or verifier lane at a given microstep can execute concurrently; expert parallelism distributes large parameter banks across devices. Halted lanes are compacted out of later batches rather than padded through more root or specialist calls. The principal systems risks are stragglers at barriers, activation imbalance, many tiny expert kernels, all-to-all communication, and route-dependent shapes that reduce batching efficiency [11, 12]. Macrocycle budgets, bounded route windows, active-lane compaction, grouped expert kernels, shared immutable prompt caches, and explicit expert-capacity limits are architectural requirements. Reported “parallelism” is measured wall-clock overlap and device utilization, not merely the existence of several logical lanes.

The model should scale only after demonstrating a favorable quality–compute frontier. A large configuration might store tens of billions of parameters while executing a root-only path for ordinary dialogue and deeper multi-path graphs for difficult coding or cross-domain work. Its identity remains the routing policy and root-preserving constraint, not a fixed active-parameter label.

8. Experimental Protocol

8.1 Falsifiable hypotheses

The proposal makes four primary predictions.

- H1.** At matched FLOPs, variable-depth graph routing improves the accuracy–compute frontier over flat fixed top- k and fixed-depth hierarchical routing.
- H2.** An always-executed root with explicit anchor constraints yields a better specialization–retention frontier than unconstrained routing or manually fixed mixture shares.
- H3.** Joint dispatch of state-isolated lanes improves verified coverage and synthesis utility over compute-matched lanes separated only by sampling noise.
- H4.** At preregistered output coverage c^* , routed verification and atomic commitment achieve lower false-commit and hard-constraint-violation rates than no-verification and immediate-publication controls while satisfying the same preregistered mean- and P95-latency budgets. Failure at c^* rejects this hypothesis; selective-risk–coverage curves are secondary analysis.

The native recurrent mechanism has five separable mechanistic predictions.

- M1.** At matched stored parameters and inference FLOPs, a tied transition with fast private, private-consolidation, and global-update timescales improves verified utility over a single-timescale recurrence and over depth-specific untied transitions.
- M2.** At matched verified task quality, adaptive private halting reduces mean and P95 recursive calls relative to fixed-depth controls, and predicted next-step gain is calibrated against paired-rollout gain.
- M3.** Route-window refresh improves specialist intervention gain relative to stage-frozen routing while remaining within preregistered switch-cost and tail-latency limits.
- M4.** Role-aligned intermediate supervision improves early-exit utility and optimization stability without reducing lane marginal utility or increasing verified redundancy.
- M5.** Barrier-only communication improves synthesis utility over no communication while preserving more useful lane diversity than per-step cross-lane attention.

8.2 Required baselines and ablations

No claim is valid from comparing only with a damaged full model. Every principal baseline must be initialized and trained independently under matched training tokens and data, with inference FLOPs, latency budgets, and parameter counts explicitly matched or reported.

Weight-tied fixed-step refinement and learned adaptive-depth refinement are also compared as recurrence-only controls [13, 14]. Multi-timescale updates, route refresh, private halting, and intermediate supervision are ablated independently; none is assumed beneficial merely from precedent.

8.3 Metrics

Evaluation must combine capability, verification, routing, and systems measurements:

- task accuracy, pass@1, exact match, repository tests, hard-constraint violations, unsupported integrations, and false commits;
- commitment coverage, selective risk, Brier score, calibration error, EVI error, verified requirement coverage, counterexample survival, and synthesis attribution;

Table 4: Minimum comparison set and the question each control answers.

System	Question
Dense shared trunk	Does conditional capacity help at all? <i>This is the decisive control for any capability claim.</i>
Flat fixed top- k MoE	Does graph structure improve over ordinary sparse routing?
Fixed-depth hierarchical MoE	Is variable stopping and expansion useful?
Connected routes with flat expert aggregation	Does the downward/upward graph sweep add value beyond using hierarchy only for selection?
Dynamic graph without private branches	Are gains due to expertise rather than parallel alternatives?
Parallel branches without specialist graph	Does hierarchical specialization add value beyond multiple canvases?
Unconstrained gate / learned gate / fixed-share sweeps	Do behavioral constraints outperform an arbitrary mixture percentage?
Causal private decoder	Does bidirectional categorical refinement improve results at matched compute?
Shared dual-mode weights vs. mode-specific adapters	Does weight sharing transfer useful representations without damaging causal behavior?
Straight-through vs. score-function router	Are routing gains robust to the biased discrete-gradient estimator?
No routed verification	Does verification predict real errors rather than repeat producer confidence?
No atomic commitment	How many incorrect blocks would immediate publication expose?
Complete HLWM	Does the full interaction improve the quality–compute frontier?

Table 5: Mechanism-specific controls for the native recurrent workspace.

Control	Question
Single-timescale recurrence	Do separate fast, private-consolidation, and global update rates add value?
Untied transition per recursive depth	Does weight tying improve parameter efficiency rather than merely reduce capacity?
Fully separate fast and slow cores	Does sharing the root-preserving kernel transfer useful computation across timescales?
Fixed-depth sweeps matched on mean and P95 FLOPs	Does adaptive private halting improve the quality–compute frontier?
Stage-frozen route / window refresh / per-step refresh	Does bounded refresh exploit new private evidence without route thrashing?
Final-only / common-target / role-aligned supervision	Does intermediate supervision improve anytime behavior without collapsing lanes?
Full / truncated gradient horizons	Are gains robust to the approximation required at scale?
Differentiable / stop-gradient summary barrier	Does cross-stage credit improve coordination or collapse private roles?
No communication / barrier-only / per-step cross-lane attention	Which schedule best balances verified synthesis and state isolation?
Terminal-only / depth-wise root anchors	Does anytime evaluation materially improve generalist retention?

- verified redundancy, lane marginal utility M_j , anchor retention at every halt-eligible depth, root/specialist residuals, and intervention effects;
- route depth and overlap, switch cost, utilization, active lanes over time, halt and next-step-gain calibration, and expert capacity violations;
- minimum, mean, P95, and maximum active parameters and FLOPs, unique parameter touch, measured overlap, latency, peak memory, and communication.

Coding evaluations should use held-out repositories and executable tests; mathematical and constraint-heavy tasks should use answer or proof checkers where available. Data contamination and benchmark memorization require explicit audits.

8.4 Statistical decision protocol

Before the final test run, the experiment preregisters primary task families, anchor margins, risk limits, output coverage c^* , compute and latency budgets, and the rule used to aggregate task utility. Example counts and independent training seeds are selected by power analysis on development data, not by stopping after a favorable run. All seeds are reported. Comparisons use paired test items where possible; confidence intervals are clustered by task or repository so correlated tokens are not treated as independent observations.

For each anchor family, root and routed non-inferiority is accepted only when the upper one-sided confidence bound on the appropriate gap in equations (17) and (22) is below its preregistered margin at every declared readout. Routing and recurrence claims compare complete quality–compute frontiers over several budgets rather than one favored depth. Lane claims use the fixed-randomness counterfactual M_j from equation (27). Halting claims compare paired next-step gains and fixed-depth sweeps matched on both mean and P95 cost. Commitment claims are evaluated at the preregistered coverage and over the full selective-risk curve. Effect sizes and uncertainty are reported even when a null hypothesis is not rejected; one seed, token-level pseudo-replication, or a post-selected threshold cannot establish an architectural advantage.

9. Experiments

The experimental program is preregistered and gated: each experiment’s battery, margins, and decision rules are frozen before its run, and each run must clear its gates before the next is authorized. Artifact-level provenance for every experiment (repository tags, checkpoints, plans, and audit records) is catalogued in Appendix A.

Experiments 1–4 share the same substrate and data boundary: the pinned Qwen/Qwen3-0.6B-Base checkpoint with frozen language weights and trainable FP32 LoRA sidecars plus HLWM modules; the Reasoning9000 synthetic release (architecture and language material only; its policy labels remain masked because no episode is independently adjudicated); and split-isolated, programmatically verified behavior anchors (arithmetic, unit conversion, ordering, missing-evidence abstention) as the only policy supervision. The graders authorize only their declared task classes. Experiment 5 keeps the substrate and widens both the data and the policy-supervision boundary with independently adjudicated and execution-verified episodes (Section 9.6).

9.1 Structural diagnostic

The original CPU prototype is a structural diagnostic, not an implementation of the complete specification. It has a shared trunk, an always-on root adapter, parent-conditioned fixed-depth routes with skip choices, four state-isolated synthetic branch roles, iterative categorical refinement, a shared-trunk verifier, and an atomic gate. It lacks the native multi-rate transition, private consolidated state, route-window refresh, adaptive per-lane depth, barrier-mediated global recurrence, role-aligned intermediate supervision, variable-depth graph traversal, discovered briefs, separately assigned verifier lanes, task-specific risk heads, EVI stopping, and retry/fallback execution.

In one diagnostic run, every configuration was trained separately for 300 updates with seed 7 and evaluated on the same 320 generated sequences (2,560 tokens). The 41,542-parameter full configura-

tion predicted 601/2,560 candidate tokens correctly (23.48%) and 0/320 complete sequences correctly. It committed 0/320 sequences, so coverage was zero and the false-commit rate conditional on commitment was undefined. The separately trained no-global-workspace control predicted 470/2,560 tokens correctly (18.36%), an observed difference of 5.12 percentage points. Several simpler controls tied or slightly exceeded the full model. These results verify old code paths only.

9.2 Experiment 1: Substrate transplant

The first GPU experiment transplanted the pinned **Qwen/Qwen3-0.6B-Base** language weights into a causal/private dual-mode backbone. It added exact categorical corruption, timestep conditioning, tied private recurrence, two isolated lanes, connected routed adapters, summary-only fan-in, autoregressive workspace-conditioned synthesis, paired clean/corrupt policy scoring, and a three-condition publication gate. The frozen Qwen substrate was adapted with FP32 LoRA sidecars in its final four blocks. This remained a reduced prototype: it used two refinement steps and six routed adapters, and it did not yet implement full macrocycle EVI, discovered lane count, separately routed verifier stages, or production serving. Table 6 reports the observed results.

Table 6: Observed Experiment 1 dual-T4 results. Behavioral metrics are from the better seed (17); stability statistics cover both seeds.

Measurement	Observed result	Interpretation
Total / trainable parameters	669,060,555 / 73,010,635	A real pretrained language substrate with parameter-efficient sidecars, not the earlier tiny network.
Hardware and replication	Two Tesla T4s; seeds 17 and 29	One independent complete seed per GPU, executed concurrently.
Training completion	896 microsteps per seed; zero skipped updates	The exact-length mixed-precision path was numerically stable for both runs.
Real-Qwen preflight	Passed before dual launch	Forward, backward, named-stage finiteness, and the 151,936-token vocabulary were exercised.
Peak allocated memory	Approximately 4.73 GB per GPU	Supports the T4 prototype regime; it is not a complete resident/reserved-memory benchmark.
Fixed-noise overfit loss	48.7% and 49.8% relative improvement	Both seeds learned the controlled local denoising signal.
Candidate vs. base token F1	0.469 vs. 0.312	The better seed improved this small held-out diagnostic; no confidence interval or external benchmark supports a superiority claim.
Prompt leakage / quality pass	0 / 0.75	The public path stopped copying prompt markers, but one quarter of checks still failed.
Narrow content accuracy	0.50	Several apparent failures were false negatives from exact-phrasal grading, but true capability remained insufficiently established.
Commitment coverage	0	Clean candidates ranked above corrupt ones, yet fixed thresholds withheld every answer.
Mean lane-summary cosine	0.947	The lanes were nearly collapsed despite different briefs.
Capability gates	Both failed	The checkpoint is not production-ready.

Two failures changed the next design. First, exact required phrases rejected semantically correct outputs such as a correct value expressed as “The answer is 36360” and a correctly sorted sequence with different prose. Second, missing-evidence anchors labeled safe abstention as “do not publish,” so a correct uncertainty statement could never become a normal public response. In addition, the clean/corrupt score ordering was learned while absolute commitment thresholds were never calibrated, and the observed lane cosine required a stronger anti-collapse signal.

The Reasoning9000 release used by these experiments contains 4,300 master episodes but no independently adjudicated episode. Its policy metadata are strongly skewed (4,277 commitments say publish) and approximately 99.5% of recorded lane routes use the same generic root–domain–specialist path. The prototype therefore masks policy, risk, halt, and verifier supervision for these rows and uses them only for architecture/language learning. Quantity alone does not repair this supervision boundary.

The trained Experiment 1 weights were not retained: the Kaggle session restarted, and the saved entry was a quick version with no model artifact. The numerical logs support the observations above, but the checkpoint cannot be reloaded for an independent rerun. This is a reproducibility failure and motivates the explicit saved-version protocol every later experiment uses, rather than reliance on ephemeral working

directories.

9.3 Experiment 2: Graded evaluation

Experiment 2 implemented the following protocol changes and was then executed as two independent, concurrent T4 seeds (Table 7); its held-out results are in Table 8.

Table 7: Experiment 2 responses to the observed Experiment 1 failures.

Observed problem	Implemented response	Required evidence
Lexical false negatives	Structured numeric, unit, ordering, and abstention graders.	Held-out accuracy by grader class, with adversarial wrong values and wording variants.
Safe abstention withheld	Treat supported missing-evidence language as a clean committed answer; use a separate fixed fallback only after policy rejection.	Safe-abstention semantic accuracy and committed-safe-response rate.
Weak policy thresholds	Fit commitment, risk, and verifier thresholds on 128 teacher-forced validation references and paired corruptions; embed them in checkpoint and adapter metadata.	Clean acceptance, corrupt rejection, balanced accuracy, and confirmation that test labels were unused.
Sparse verified supervision	Expand split-isolated anchors to 512 train, 128 validation, and 128 test records, balanced across four semantic task classes.	Zero prompt overlap across splits and independent checks that every clean/negative pair is graded correctly.
Lane collapse	Apply squared-cosine diversity loss in both single-pass local and joint phases, with stronger weight, while retaining role-specific briefs.	Lower absolute held-out lane cosine plus unchanged or improved task quality and lane marginal utility.
Small output audit	Evaluate 32 held-out generations per seed (64 across seeds), half narrow probes and half Reasoning9000 episodes.	Per-seed reports, cross-seed gate, full reverse timesteps, routing load, leakage, semantic quality, and F1 controls.
Artifact loss	Require <i>Save & Run All</i> ; create a best-adapter archive and a separate resumable checkpoint with SHA-256 sidecars and explicit download links.	Saved-version output existence, byte count, checksum verification, and successful fresh-process reload.

Table 8: Observed Experiment 2 held-out results. Each seed was evaluated on 16 behavior anchors and 16 Reasoning9000 records.

Measurement	Seed 17	Seed 29	Interpretation
Completed microsteps / skipped updates	1,024 / 0	1,024 / 0	Both optimization runs completed stably.
Peak allocated memory	4.732 GB	4.733 GB	The longer run preserved the T4 memory regime.
Fixed-noise overfit improvement	46.87%	47.73%	Both seeds learned the controlled denoising slice.
Candidate / frozen-base token F1	0.390 / 0.288	0.421 / 0.288	Candidate F1 improved on this small diagnostic, without an external superiority claim.
Quality pass / prompt leakage	0.750 / 0	0.781 / 0	Surface quality improved and no prompt markers leaked.
Probe semantic accuracy	0.625	0.563	Neither seed reached the preregistered 0.75 gate.
Safe-abstention accuracy	0	1.000	The behavior was highly seed-sensitive.
Probe commitment accuracy	0.625	0.313	Correct generated candidates were frequently withheld and incorrect ones were not separated reliably.
Commitment coverage	0.125	0.125	Only four of 32 candidates were published per seed.
Mean lane-summary cosine	0.496	0.331	The stronger diversity objective materially reduced the earlier collapse diagnostic.
Capability gate	Failed	Failed	Experiment 2 is not production-ready.

The calibration report appeared strong in isolation: on teacher-forced clean references and their paired corruptions, seed 29 obtained 0.906 clean acceptance, 0.930 corrupt rejection, and 0.918 balanced accuracy without consulting the test split. That estimate did not describe the deployed decision problem. Freely generated candidates changed both the candidate representations and the private verifier distribution. For example, correct missing-evidence answers received candidate commitment probabilities near 0.95 but private denoising-error probabilities near 0.74, above the fitted 0.590 threshold, and were withheld. This is a calibration-to-inference mismatch, not evidence that supported abstention should remain private.

9.4 Experiment 3: Publication boundary

Experiment 3 made two architectural corrections. First, the commitment module became candidate-level: it emits three candidate-level logits (publish, risk and candidate error), and the exact frozen candidate snapshot supplies all three scores. Private denoising error remains a diagnostic and training signal but is no longer a hard proxy for final-answer validity. Second, threshold fitting runs the complete generator on validation anchors, labels those emissions with deterministic semantic graders, adds mechanically corrupted snapshots as negatives, and jointly searches all three thresholds. The test split remains untouched. Experiment 3 also expanded multi-template anchors to 1,024 train, 256 validation, and 256 test records, used matched operand distributions across isolated splits, increased each seed to 4,224 microsteps with wider LoRA coverage, and doubled the held-out audit to 64 outputs per seed. Table 9 reports the results.

Table 9: Observed Experiment 3 held-out results. Each seed was evaluated on 32 behavior anchors and 32 Reasoning9000 records.

Measurement	Seed 17	Seed 29	Interpretation
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	Both longer optimization runs completed stably.
Peak allocated memory	4.769 GB	4.770 GB	Wider rank-16 adapters retained the dual-T4 memory regime.
Fixed-noise overfit improvement	54.58%	55.97%	Both seeds learned the controlled denoising slice.
Candidate / frozen-base token F1	0.527 / 0.253	0.536 / 0.253	Both seeds more than doubled this small token-overlap diagnostic.
Quality pass / prompt leakage	0.813 / 0	0.891 / 0	Both audits cleared surface quality and leakage gates.
Probe semantic accuracy	0.625	0.813	Only seed 29 cleared the 0.75 semantic gate.
Safe-abstention accuracy / committed rate	1.000 / 1.000	0.875 / 0.875	The public safe-abstention target worked in both seeds.
Probe commitment accuracy	0.625	0.750	Only seed 29 cleared the held-out commitment gate.
Mean lane-summary cosine	0.707	0.805	The formal lane-distinctness gate passed, but seed stability remains unresolved.
Validation generated-emission calibration	Failed	0.839 positive accept; 0.583 negative reject	Candidate scoring did not separate generated hard negatives at the required 0.70 rate.
Capability gate	Failed	Failed	Neither checkpoint authorizes the next benchmark or production use.

Experiment 3 therefore fixed the earlier private-denoising publication error and substantially improved the selected seed’s narrow behavior, but it did not fix policy-score separation or cross-seed robustness. Seed 29’s validation emissions were semantically correct at 0.875 and its fitted thresholds achieved 0.839 positive acceptance, yet negative rejection was only 0.583 and balanced accuracy was 0.711. The threshold search could not manufacture a missing representation margin: a commitment threshold of zero and risk/error thresholds near 10^{-3} are evidence of overlapping, poorly scaled policy scores rather than a deployable operating point. Routing also remained narrow: seed 17 sent 98.4% of audited decisions through one expert and seed 29 used only two experts materially, so the then-current “multiple routes” gate was too weak to demonstrate specialist use.

The next controlled experiment was designed against exactly these two failures, with a preregistered plan: an on-policy policy-head phase on generated *training*-anchor emissions with programmatic semantic labels, invalid emissions, and harder meaning-changing corruptions, with thresholds fitted only afterward on generated validation emissions and tested once on the isolated test split; a router marginal-entropy regularizer; and preregistered entropy, minimum-load, and intervention-based routing gates rather than counting any second route above one percent.

9.5 Experiment 4: On-policy calibration and two-seed replication

Design. Experiment 4 keeps Experiment 3’s architecture and adds: (i) an *on-policy policy-head phase*, in which, after joint training, the frozen decision path generates 96 train-anchor emissions, deterministic graders label them, and only the three candidate-level heads are fitted on those emissions plus two hard-negative families (the anchor’s fluent wrong-value answer and a mechanically corrupted snapshot); thresholds are then fitted, as before, on 64 generated *validation* emissions, and the test split remains untouched; (ii) a *router marginal-entropy regularizer* (KL divergence of mean routing probabilities to uniform, weight 0.02) alongside the existing balance term; and (iii) three preregistered routing gates: second-route load ≥ 0.10 , normalized route entropy ≥ 0.25 , and a causal intervention that pins every route to the least-used expert and must move mean absolute commitment probability by ≥ 0.01 on identical frozen candidates. Two independent seeds (17 and 29) trained concurrently, one per T4, 4,224 microsteps each with zero skipped updates permitted, followed by per-seed 64-output fresh-process audits. The preregistered primary endpoint: *both seeds pass the complete gate*.

Results. The endpoint failed, in the most informative way available: the two seeds agree exactly on what works and what does not (Table 10).

Table 10: Observed Experiment 4 per-seed results, evaluated on 64 fresh-process outputs per seed. Failing entries are marked in red.

Measurement (per seed)	Seed 17	Seed 29	Gate
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	exact / 0
Peak allocated memory (GB)	4.77	4.77	n/a
Prompt leakage	0.0	0.0	= 0
Quality pass rate	0.938	0.906	≥ 0.75
Probe semantic accuracy	0.875	0.844	≥ 0.75
Safe-abstention accuracy / committed rate	1.00 / 1.00	1.00 / 1.00	≥ 0.75 / 0.70
Probe commitment accuracy	0.844	0.906	≥ 0.70
Candidate token F1 (frozen base 0.273)	0.571	0.505	$\geq 0.8 \times \text{base}$
Lane summary cosine	0.860	0.831	< 0.90
<i>Calibration</i> : positive accept	1.000	0.982	≥ 0.70
<i>Calibration</i> : negative reject (Experiment 3: 0.583)	0.889	0.917	≥ 0.70
<i>Calibration</i> : balanced accuracy	0.944	0.949	≥ 0.70
Second-route load	0.063	0.094	≥ 0.10
Normalized route entropy	0.130	0.174	≥ 0.25
Routing intervention $ \Delta \text{commit} $	0.0035	0.0008	≥ 0.01
Complete capability gate	Failed	Failed	n/a

The calibration failure of Experiment 3 is fixed and the fix replicates. Held-out negative rejection rose from 0.583 to 0.889 and 0.917 on seeds trained independently, with balanced accuracy near 0.95 on both; every behavioral, leakage, quality, margin, and lane gate passed on both seeds. The improvement held on *held-out* generated validation emissions, so this is not the heads memorizing their 288 training records. The on-policy head phase did precisely what it was designed to do.

The routing mechanism failed its gates identically on both seeds, and the failure is now mechanistically understood. Across all lane-and-depth routing decisions in the audits, expert 0 received 93.8% (seed 17) and 90.6% (seed 29) of traffic, a second expert received the remainder, and experts 2–5 received *zero* decisions on both seeds. Training telemetry shows the marginal-entropy regularizer did its literal job, driving the KL divergence of mean routing probabilities to uniform down to 0.01–0.03, while hard decisions stayed concentrated: the regularizer moved the probabilities, not the choices. Its side effect

is worse: near-uniform *soft* mixing feeds every expert nearly identical gradients, homogenizing them, which the causal intervention then confirms: pinning all routing to the least-used expert moves mean absolute commitment probability by only 0.0035/0.0008 against the 0.01 gate. Route diversity that survives is not causally load-bearing. Relative to Experiment 3 the one-expert share did fall (98.4% $\rightarrow$ 90.6–93.8%) and second-route load roughly doubled; the direction is right, the magnitude fails, and the mechanism is wrong.

Secondary observations, recorded against over-reading the headline calibration numbers.

First, the policy heads are *saturated*: after 400 head-phase epochs, mean negative commitment probability sits near 5×10^{-5} and positive near 0.9999, and most audit commitment probabilities lie at the rails. Ranking accuracy of 1.0 on head-phase data is memorization; the honest figures are the held-out 0.889/0.917. Second, the fitted commitment threshold is 0.0 on both seeds, because the joint threshold search effectively disabled the commitment head and assigned rejection to the risk and verifier thresholds (0.973/0.973 on seed 17; 0.883/0.762 on seed 29). A three-head gate whose calibrated policy uses two heads is carrying dead weight; Section 11 explains the correlated label structure that predicts exactly this. Third, gate agreement does not imply behavioral identity: at their fitted operating points seed 17 committed on 75% of audit outputs (commit accuracy 0.719) and seed 29 on 41% (0.453). Selective-prediction reporting must therefore always pair coverage with risk, per seed.

Decisions taken. Following the preregistered rules: additional seeds of this revision are not run (two seeds already replicate the failure); the external-benchmark step is *not* authorized, because the full gate failed; and the next revision changes the routing objective to load balancing on *hard assignments* (a Switch-style auxiliary loss on realized route counts [12]) in place of the marginal-KL term, plus an expert-output diversity penalty so experts cannot satisfy the objective by converging, with a declared fallback of reducing the expert count from six to two if diversity still fails to emerge. The head phase is additionally shortened or label-smoothed so calibration operates on an unsaturated score scale, which also unconfounds the causal-liveness measurement.

9.6 Experiment 5: Routing redesign on verified multi-domain data

Design. Experiment 5 attacks the two Experiment 4 verdicts directly. First, the routing objective is replaced: the marginal-entropy regularizer is removed, the Switch-style balance loss on hard assignments is strengthened (weight 0.01 $\rightarrow$ 0.05), and a pairwise expert-output diversity penalty (weight 0.05) evaluates every expert on shared probe rows so decorrelation pressure does not depend on routing traffic. Second, the policy heads are de-saturated: 120 head-phase epochs (from 400) with 0.05 label smoothing. Third, the data boundary widens substantially: alongside Reasoning9000 and the behavior anchors, training adds independently adjudicated builder episodes (verified by executed pytest), *execution-verified* benchmark train-split episodes from MBPP, APPS, TACO and CodeContests, problems whose official reference solution passed its own official checks in an isolated subprocess at dataset-build time (1,699 episodes; 166 references that failed their own checks were dropped rather than shipped unverified), plus Spider text-to-SQL and audited reasoning packets as causal-only material, under cross-split collision and duplicate drops. Official evaluation splits of every benchmark remain untouched. Token budgets grow to context 256 / canvas 128 / causal 1,024 so the code data fits; audits are domain-stratified with execution and SQL graders (per-domain metrics report-only). Gate thresholds are unchanged from the Experiment 4 preregistration. Two independent seeds (17, 29) trained 4,224 microsteps each, one per T4.

Results. The primary endpoint (both seeds pass) failed, but for the first time in the program the seeds did not fail the same way (Table 11).

The redesigned routing objective can produce real route diversity. Seed 29 routed through three experts (loads 0.16/0.42/0.42), passing second-route load at $4.2\times$ its gate and normalized entropy at $2.3\times$ its gate, thresholds that Experiments 2–4 never exceeded 0.094 and 0.174 on. The causal-liveness intervention improved to 0.0073 against the 0.01 gate (Experiment 4: 0.0008–0.0035) but still fell short. Notably, telemetry shows the expert-output diversity penalty itself never engaged: the expert adapters’ zero-initialized output projections keep all expert deltas at zero early in training, so there is nothing to decorrelate and the term stays inert. The diversity seed 29 found is attributable to the strengthened hard-assignment balance term plus favorable optimization noise.

The same configuration produced a catastrophically different sibling. Seed 17 collapsed to

Table 11: Observed Experiment 5 per-seed results, evaluated on 64 fresh-process domain-stratified outputs per seed. Failing entries are marked in red; the first-ever routing-load passes are in bold.

Measurement (per seed)	Seed 17	Seed 29	Gate
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	exact / 0
Peak allocated memory (GB)	4.82	4.82	n/a
Prompt leakage	0.266	0.0	= 0
Quality pass rate	0.578	0.844	≥ 0.75
Probe semantic accuracy	0.281	0.813	≥ 0.75
Safe-abstention accuracy / committed rate	0.75 / 0.75	1.00 / 1.00	≥ 0.75 / 0.70
Probe commit accuracy	0.656	0.625	≥ 0.70
Candidate token F1 (frozen base 0.307)	0.324	0.585	$\geq 0.8 \times$ base
Causal-path token F1 (diagnostic)	0.588	0.589	n/a
Lane summary cosine	0.942	0.890	< 0.90
Calibration: accept / reject / balanced	0.818 / 1.000 / 0.909	0.875 / 0.933 / 0.904	≥ 0.70
Second-route load	0.000	0.418	≥ 0.10
Normalized route entropy	0.000	0.570	≥ 0.25
Routing intervention $ \Delta$ commit	0.0035	0.0073	≥ 0.01
Complete capability gate	Failed (8 of 18)	Failed (2 of 18)	n/a

100% one-expert routing (worse than any prior experiment), lane cosine 0.942, and a degenerate candidate path: 26.6% prompt-leak flags and 0.281 anchor accuracy. The failure texture is informative: candidates are frequently *numerically correct but format-broken* (e.g. reference “The result is 84325.” versus candidate “Human: 80329 + 3996 = 84325.”), the plain causal path is completely healthy (token F1 0.588, identical to seed 29’s), and only 3 of 9,284 training rows contain such chat markers, so this is not data contamination but a degenerate workspace prefix that stopped steering generation, letting the base model’s dialogue prior leak through. Every training-time signal looked normal (calibration 0.818/1.000, non-degenerate thresholds, smooth losses): the collapse is visible only at generation time. The joint workspace phase therefore has high, currently undetectable training variance.

Head de-saturation worked on both seeds. Calibrated scores now sit at 0.05/0.95 rather than Experiment 4’s 5×10^{-5} /0.9999 rails, with balanced accuracy ≈ 0.90 on held-out generated emissions. Seed 29’s failing commit accuracy decomposes cleanly: 11 of its 12 wrong decisions are confident false *rejections* of valid answers whose head scores sit exactly at the smoothed negative target: a cluster of valid candidates (the ordering family is weakest) that the 96-record head phase evidently under-represented as positives. The commitment threshold also degenerated to zero again on seed 29, the third recurrence of the correlated-label symptom.

Decisions taken. Benchmarks remain unauthorized (full gate required). The declared fallback of cutting the expert count is *not* triggered, since it required both seeds to fail the routing gates, and seed 29 passed two of three. The next revision changes exactly two things, both aimed at mechanisms this experiment identified: expert output projections are initialized at a small nonzero scale so the diversity penalty is active from the first step (targeting the seed-17 collapse mode), and the head phase grows to 128 records with per-family round-robin balancing (targeting the false-rejection cluster). Everything else is frozen so the variance question receives a clean second measurement on the same seeds.

9.7 Experiment 6: Targeted variance fixes on frozen configuration

Design. Experiment 6 is the clean second measurement that Experiment 5’s decisions demanded. Exactly two flags change: expert output projections are initialized at standard deviation 0.01 instead of zero (so the expert-output diversity penalty, provably inert throughout Experiment 5, exerts pressure from the first step, the lever aimed at seed 17’s collapse), and the head phase grows from 96 to 128 records with per-family round-robin balancing (the lever aimed at seed 29’s cluster of confident false rejections). Data (byte-identical), budgets, balance weight, label smoothing, steps (4,224), gate thresholds, and audit design are frozen. The same two seeds are deliberately re-run: the preregistered question is whether the fixes rescue seed 17’s exact failure mode and lift seed 29’s one remaining behavioral gate.

Results. The primary endpoint failed on both seeds, but the two fixes had cleanly separable fates (Table 12).

Table 12: Observed Experiment 6 per-seed results (same protocol as Table 11); Experiment 5 values in parentheses where they changed materially. Failing entries in red; rescued entries in bold.

Measurement (per seed)	Seed 17	Seed 29	Gate
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	exact / 0
Prompt leakage	0.0 (0.266)	0.0	= 0
Probe semantic accuracy	0.781 (0.281)	0.750	≥ 0.75
Safe-abstention accuracy / committed rate	1.00 / 1.00	0.875 / 0.875	≥ 0.75 / 0.70
Probe commit accuracy	0.781 (0.656)	0.625 (0.625)	≥ 0.70
Lane summary cosine	0.902 (0.942)	0.919 (0.890)	< 0.90
Second-route load	0.000	0.113 (0.418)	≥ 0.10
Normalized route entropy	0.000	0.306 (0.570)	≥ 0.25
Routing intervention $ \Delta \text{commit} $	0.0012	0.0031	≥ 0.01
Fitted commitment threshold	0.0 (degenerate)	0.090	non-deg. (diagnostic)
Complete capability gate	Failed (4 of 18)	Failed (4 of 18)	n/a

Nonzero expert initialization rescued the degeneration cluster, and only that. Seed 17’s Experiment 5 pathology (26.6% prompt leak, 0.281 anchor accuracy, dialogue-prior leakage) is gone: zero leakage, probe accuracy 0.781, and commit accuracy 0.781 passing its gate for the first time on this seed. Yet the same seed’s routing collapsed to a single expert again, with second-route load and entropy exactly zero under a different initialization. Prefix degeneration and routing collapse were therefore *not one failure mode*; the initialization fix decoupled them, and the collapse mode is now 2-for-2 on this seed across two initializations.

The head-phase balancing fix was nullified before it could act. Seed 29’s commit accuracy is unchanged to the third decimal (0.625): of 128 requested head-phase emissions, 85–87 were graded invalid at collection time, leaving 41–43 valid positives against ≈ 342 negatives, the same $\approx 1:8$ imbalance the fix targeted. Round-robin balanced the *requests*, not the *valid records*. The corrective is mechanical and preregistered for the next experiment: enforce a per-family floor of valid positives after grader filtering, resampling until met.

Routing diversity is fragile, not merely seed-dependent. The seed that had passed the load gates at $4.2\times$ and $2.3\times$ margin now passes them at $1.1\times$ and $1.2\times$ (load $0.418 \rightarrow 0.113$, entropy $0.570 \rightarrow 0.306$), the causal-liveness gate failed on both seeds for the fourth consecutive experiment (now 0-for-8 seed-runs), and the degenerate commitment threshold recurred for the fourth time, migrating to seed 17, whose Experiment 5 threshold had been healthy. One further negative result is methodological: the uploaded run telemetry resumed past the training window, so whether the diversity penalty actually engaged under nonzero initialization is *unmeasured, not answered*; the next experiment must ship complete telemetry.

Decisions taken. Benchmarks remain unauthorized. The preregistered fallback ladder’s second rung is now triggered: joint-phase outcomes remained a seed lottery *after* the targeted fixes, which is the exact condition under which Table 18 prescribes simplifying the workspace, and the routing-collapse row independently prescribes cutting the expert count. The next experiment is therefore a *reduction experiment*, built and preregistered with frozen parameters: expert count $6 \rightarrow 2$ with nonzero expert initialization retained; validity-gated family balancing in the head phase (collection continues past the 128-record budget until every anchor family holds at least 16 grader-verified valid positives, capped at 512 attempts, with a shortfall reported rather than fatal); a report-only generation-time canary at every 512-step evaluation boundary, being eight stratified anchors scored for prompt leakage, dialogue-marker drift, semantic validity, and realized route entropy, written to the training log (both prior collapses were invisible to every training-time signal); and a declared threshold non-degeneracy diagnostic (calibrated commitment threshold within $[0.02, 0.98]$), with everything else frozen and the same seeds. If either seed still collapses at two experts, the expert graph is removed from the small-scale claim entirely and the routing section of this paper is re-scoped to proposal-plus-negative-evidence. If the full gate passes on both seeds, the matched plain-LoRA control runs immediately, before any benchmark.

9.8 Experiment 7: Minimum-viable routing

Design. Experiment 7 executes the ladder rung Experiment 6 invoked, with the mechanism-level repairs Experiment 6 identified. Four changes against the frozen configuration: expert count $6 \rightarrow 2$ (nonzero initialization retained); a validity-gated head phase (collection continues past the 128-record budget until every anchor family holds at least 16 grader-verified valid positives, capped at 512 attempts, shortfall reported rather than fatal); a report-only generation-time canary at every 512-step evaluation boundary (eight stratified anchors scored for leakage, dialogue-marker drift, semantic validity, and realized route entropy); and the declared threshold non-degeneracy diagnostic. Data, budgets, steps, gate thresholds, audit design, and seeds are frozen. Training telemetry is complete for the first time since Experiment 5.

Results. The primary endpoint failed on both seeds (6 and 5 gates respectively), and the preregistered rung-1 condition, “if either seed still collapses at two experts,” was met (Table 13).

Table 13: Observed Experiment 7 per-seed results; Experiment 6 values in parentheses where they changed materially. Failing entries in red; program-best entries in bold.

Measurement (per seed)	Seed 17	Seed 29	Gate
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	exact / 0
Prompt leakage	0.0	0.0625 (0.0)	= 0
Probe semantic accuracy	0.563 (0.781)	0.469 (0.750)	≥ 0.75
Probe commit accuracy	0.844 (0.781)	0.469 (0.625)	≥ 0.70
Lane summary cosine	0.928 (0.902)	0.829 (0.919)	< 0.90
Second-route load	0.000	0.273 (0.113)	≥ 0.10
Normalized route entropy	0.000	0.846 (0.306)	≥ 0.25
Routing intervention $ \Delta \text{commit} $	0.0082 (0.0012)	0.0037	≥ 0.01
Head-phase valid positives (balanced)	78 of 512 attempts	77 of 448	floor 16/family
Fitted commitment threshold	0.0 (degenerate)	0.0 (degenerate)	non-deg. (diagnostic)
Complete capability gate	Failed (6 of 18)	Failed (5 of 18)	n/a

The reduction is a routing/content trade, not a fix. At two experts the non-collapsing seed posted the healthiest routing in the program, with second-route load 0.273, normalized entropy 0.846 and the lane-distinctness gate passing for the first time in seven experiments, while both seeds lost 0.19–0.28 of probe content accuracy and both numeric and unit probes fell to 0.25 and 0.125. The six-expert bank had been carrying *content* capacity even while failing every causal-routing gate; shrinking it degraded the generator. Meanwhile seed 17 collapsed to a single expert for the third time under a third configuration.

Validity-gated balancing worked, and moved the head gate where content held. The floor logic produced 77–78 valid positives balanced across families (versus 41–43 unbalanced in Experiment 6), honestly reporting the unit family short at its measured 0.05–0.08 grader accuracy. Seed 17’s commit accuracy then passed at 0.844, the best measured in the program, direct evidence that Experiment 6’s head-phase imbalance was causal for the confident-false-rejection cluster. The commitment threshold nonetheless calibrated degenerate on both seeds (recurrences five and six): label de-correlation is now mandatory, not scheduled.

The canary’s first finding is about the canary. Complete telemetry finally answers Experiment 6’s open question: the expert-diversity penalty engages under nonzero initialization (growing two orders of magnitude during training) but saturates far below the weight’s bite point. On routing, however, the anchor-stratified canary was blind by construction: both seeds’ canaries showed single-expert routing at every boundary, including the seed whose audit measured 0.273 second-route load. Behavior anchors route homogeneously, and route diversity lives in the non-anchor domains the canary never sampled. A useful canary must be domain-stratified.

Decisions taken. Benchmarks remain unauthorized. Rung 1 of the preregistered ladder is executed: the routed expert graph exits the small-scale claim, the follow-up runs workspace-only (lanes, synthesis, verification and calibrated commitment, with the freed trainable budget returned to the LoRA sidecars), and this paper’s routing sections stand as proposal plus negative evidence (three collapses under three configurations; load gates passed by one seed at six experts twice and at two experts once; causal liveness 0-for-10 seed-runs). The successor configuration carries forward the validity-gated head phase, replaces

the anchor-only canary with a domain-stratified one, and must ship the policy-label de-correlation fix. Its content question is declared in advance: if workspace-only at matched trainable budget recovers Experiment 6 content levels, the expert bank’s content contribution was fungible capacity; if not, the reduction claim carries that caveat.

9.9 Experiment 8: Verified fan-in

Design (frozen before the run). Experiment 8 ran the workspace-only configuration Experiment 7 authorized: one expert, two lanes, the freed trainable budget returned to the LoRA sidecars, the validity-gated head phase carried forward, and a domain-stratified training canary replacing the anchor-only one. Three preregistered mechanisms were added, each shipping with its own causal test or killer baseline.

M1. Latent read-out memory. The synthesis prefix had been a linear reshape of exactly three pooled vectors into eight positions (Known Limitations item 2), so lane computation was information-bottlenecked out of the public answer regardless of what the lanes learned. Experiment 8 widened it to 42 positions: the original eight pooled tokens plus 34 memory tokens, being sixteen windowed masked means of each lane’s final canvas hidden states (32 tokens) and one summary token per lane. The memory passed through zero-initialized, input-normalized projections, so the channel opened only as gradients demanded. The synchronization-barrier invariant held: hidden states cross, token identities never do. Every audit row was re-decoded with the memory sliced off, so the widened channel’s causal contribution was measured in the same run.

M2. Verified fan-in. The calibrated heads, the one replicated mechanism in the program, had only ever braked; Experiment 8 made them steer. One workspace pass decoded four candidates (greedy plus temperatures 0.7, 0.9 and 1.1); the heads scored each through the identical private-state feature path used at head training; the best-scoring candidate was published if it cleared the fitted rule. Head-phase collection harvested all four candidates per anchor with within-anchor ranking pairs. The audit spent the same four-candidate budget on the plain causal path and reported self-consistency majority voting and maximum mean-log-probability selection on the same rows. The declared headline gate was fan-in beating self-consistency at matched candidate budget.

M3. Scalar publication rule. Six rail-degenerate thresholds under the joint triple-threshold sweep were answered structurally: a regularized logistic combiner over the three de-correlated head probabilities, with a single threshold placed at the margin midpoint nearest one half, fitted on N-best validation candidates and doing double duty as selector score and abstention gate. A coverage-matched mean-log-probability abstention baseline, given a post-hoc threshold in its own favor, was reported alongside it.

The battery grew from 15 to 20 gates. Seeds, data, budgets, the pinned base revision and the 4,224 microsteps were frozen. The run executed on two T4 GPUs in bf16, one seed per device, with a 64-row audit per seed.

Results. The replication failed on both seeds: nine of twenty gates failed on seed 17 and eleven of twenty on seed 29 (Table 14). The headline gate failed decisively and in the same direction on both seeds, while the preregistered rung-2 condition was met exactly: fan-in was no worse than its own greedy baseline on both seeds and still lost to self-consistency.

The selector was near-oracle; the generation channel starved it. The loss to self-consistency was not a selection failure. On the 36 graded rows, seed 17’s publish-score selection reached its own oracle exactly (selected 0.417 against an any-valid oracle of 0.417, from a greedy baseline of 0.194), and seed 29 recovered half the available headroom (0.472 against an oracle of 0.556, from 0.444). The problem was the pool it selected from. The workspace-conditioned oracle, 0.417 and 0.556, sat roughly thirty points below plain causal greedy decoding from the *same adapter on the same rows*, 0.758 and 0.818. Token F1 tells the same story: 0.289 and 0.454 for workspace candidates against 0.560 and 0.570 for causal ones. Self-consistency voting, maximum-log-probability best-of- N and causal greedy decoding all tied at 0.758 and 0.818 because all three drew from the healthy causal channel, while fan-in drew from the degraded workspace channel. No selection rule recovers content that was never generated.

The read-out memory was live, which sharpens rather than softens the failure. The pre-registered null-ablation rung was cleanly rejected: zeroing the 34 memory tokens cost 0.167 and 0.306 of

Table 14: Observed Experiment 8 per-seed results. Failing entries in red; program-first results in bold. Audit $n = 64$ published rows per seed, of which 36 carry an authorized grader; the baseline comparison uses the 33 rows on which all four arms produced a light-gradable answer, so its denominator differs from the fan-in block’s by three rows.

Measurement (per seed)	Seed 17	Seed 29	Gate
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	exact / 0
Validation calibration balanced accuracy	0.884	0.877	≥ 0.70
Fitted publish threshold	0.578	0.689	non-degenerate
Turn-scaffold leak rate	0.188	0.016	$= 0$
Quality pass rate	0.563	0.688	≥ 0.75
Probe content accuracy	0.250	0.500	≥ 0.75
Probe commit accuracy	0.656	0.406	≥ 0.70
Safe-abstention accuracy	0.625	0.000	≥ 0.75
Lane summary cosine	0.945	0.968	< 0.90
Workspace token F1 / causal token F1	0.289 / 0.560	0.454 / 0.570	report
Read-out ablation Δ graded accuracy	0.167	0.306	≥ 0.05 (non-null)
Fan-in selected vs. its own greedy	+0.222	+0.028	≥ 0
Fan-in selected vs. self-consistency	0.455 vs. 0.758	0.515 vs. 0.818	fan-in $>$ SC
Selective accuracy, heads vs. log-probability	0.818 vs. 0.636	0.333 vs. 1.000	heads $\geq$ logprob
at matched coverage	(0.306)	(0.083)	report
Complete capability gate	Failed (9 of 20)	Failed (11 of 20)	n/a

graded accuracy and 0.104 and 0.232 of token F1. The lanes did inform the answer. The channel therefore carried real information and simultaneously corrupted the generation register, which is the most useful diagnostic the program has produced. The negative result is “conditioning on the workspace prefix costs more than the workspace knows,” not “the workspace knows nothing.”

The corruption was visible in the emissions. Session D crashed before reaching the audit, so this was the program’s first look at the workspace emission path. Twelve of 64 seed-17 published answers and one of 64 on seed 29 opened with dialogue scaffold: "Human: 46920 seconds.", "Assistant: The software version was not provided...". The same episodes’ causal generations were clean, "The result is 84325." against the workspace channel’s "Human: 84325.". The domain-stratified training canary reported a leak rate of 0.0 at every checkpoint on both seeds, so the pathology was specific to the audit emission path and invisible to training-time telemetry. The leak is nonetheless a symptom and not the binding failure: counterfactually forgiving every quality failure whose only defect was scaffold lifts the pass rates to 0.609 and 0.703, still short of the 0.75 gate. Semantic content loss is the main event and the scaffold is its visible edge.

The scalar publication rule held, and one seed’s operating point strangled coverage. The six-run degenerate-threshold streak was broken: thresholds fitted at 0.578 and 0.689, head separation was healthy (mean commit probability 0.80 on positives against 0.05 on negatives, pairwise ranking 0.999 and 0.998), and calibration balanced accuracy reached 0.884 and 0.877. A calibrated rule is still only as good as where its operating point lands. Seed 29’s fitted rule, weights $[2.52, -1.80, -1.00]$ with $\tau = 0.689$, committed on 7.8% of audit rows and on none of the abstention battery’s commit-expected probes; its safe-abstention accuracy of 0.000 is over-conservatism rather than miscalibration, since its corrupt rejection rate was 0.922. Seed 17 committed on 17.2% of rows and posted the program’s first selective-prediction win over a matched-coverage log-probability baseline, 0.818 against 0.636 at coverage 0.306 (Figure 5).

Smaller negatives, disclosed. Lane summaries were not distinct on either seed (cosine 0.945 and 0.968), so Experiment 7’s single distinct-lane result did not carry into the workspace-only architecture. Seed 29’s verifier head ranked corrupt candidates *below* clean ones on the audit (margin -0.109) despite ranking them correctly at policy-fit time. All baseline comparisons rest on 33 to 36 graded rows per seed, as the preregistration disclosed in advance.

Why the headline gate was hard to win on evidence. A post-hoc analysis conducted while designing Experiment 9, and therefore not part of the preregistration, shows the gate was also badly placed. Let p be the probability that a single sampled candidate is correct. The canonical row on which a perfect selector wins and plurality voting loses is the row where exactly one of N candidates is correct, which occurs with probability $Np(1-p)^{N-1}$ (Figure 4). Experiment 8’s self-consistency baseline operated on the

causal pool at p between 0.758 and 0.818, where that quantity is 0.043 down to 0.020 at $N = 4$: between one row in 23 and one row in 50, against a 33-row graded audit. Even with a repaired channel and a perfect selector, the comparison could not have produced a decidable result at this sample size. Experiment 8’s headline was thus doubly compromised, once by the channel and once by the arithmetic of the comparison, and Experiment 9 separates both.

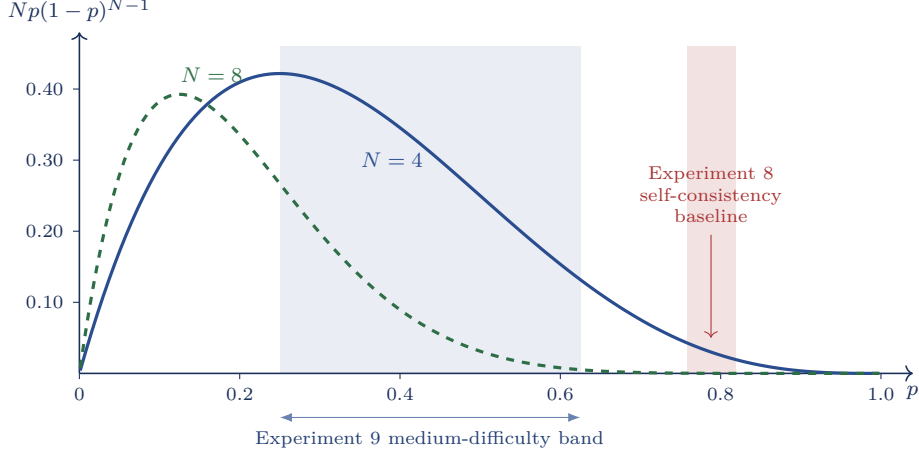


Figure 4: Where a selector can beat plurality voting. The curves plot $Np(1-p)^{N-1}$, the probability that exactly one of N independent candidates is correct, which is the canonical row a perfect selector wins and majority voting loses. Experiment 8 placed its headline comparison against a self-consistency baseline running at $p \in [0.758, 0.818]$ (red band), where the quantity is 0.043 to 0.020 at $N = 4$, one or two rows of a 33-row audit. Experiment 9 moves the headline onto the model-estimated medium-difficulty band $p \in [0.25, 0.625]$ (blue band), where at $N = 8$ it runs from 0.267 at $p = 0.25$ to 0.005 at $p = 0.625$. The room is real but concentrated at the lower end of the band, which is why Experiment 9 also reports every difficulty bin and a pooled test rather than the headline alone.

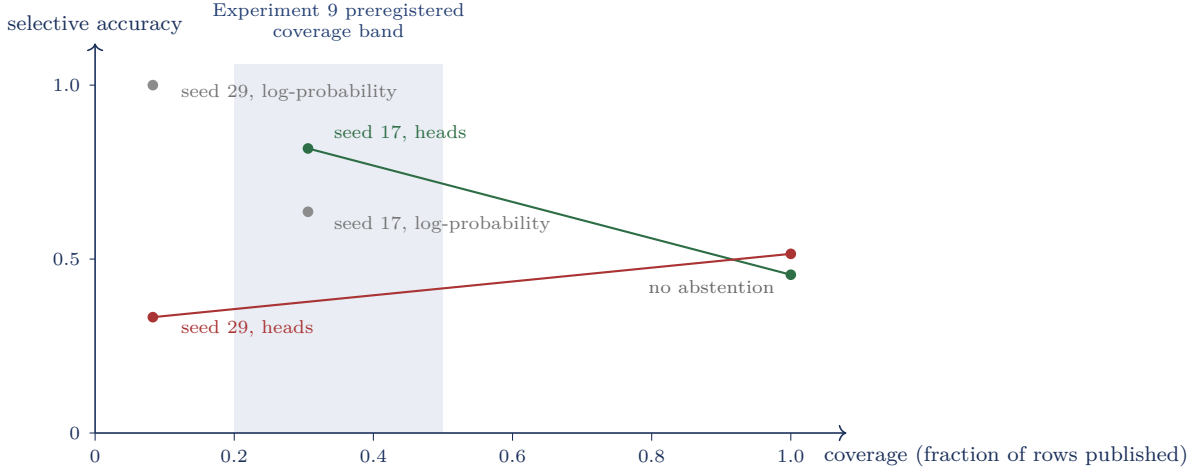


Figure 5: Experiment 8 selective prediction, both seeds. Each colored line joins a seed’s published operating point to its own no-abstention accuracy; the gray points are the log-probability baseline evaluated at the same coverage. Seed 17 abstained at coverage 0.306 and gained 18 points over the baseline, the program’s first selective-prediction win. Seed 29’s rule landed at coverage 0.083, three published rows, where its accuracy of 0.333 loses to a baseline that happens to be perfect on those three rows; that seed’s safe-abstention accuracy of 0.000 is over-conservatism, not miscalibration. The claim is therefore supported on one seed of two and is not replicated. Experiment 9 replaces that threshold rule with a split-conformal order statistic whose realized coverage is targeted into the shaded band by construction, which removes the failure mode seed 29 exhibits here.

Decisions taken. Rung 2 of the preregistered fallback ladder is invoked. The verified fan-in claim is dead at this scale: the mechanism joins routing as proposal plus negative evidence, and the paper’s claim

re-scopes to calibrated abstention alone, with the seed split disclosed as observed rather than averaged away. Rungs 3 and 4 explicitly did not fire, because the read-out ablation was non-null on both seeds and the scalar publish rule stayed non-degenerate on both; those two structural fixes are confirmed working and carry forward. The matched plain-LoRA control and the external benchmarks were gated behind a full gate pass and therefore remain unauthorized. Experiment 8’s other contribution is procedural: seed 17’s commitment calibration and seed 29’s policy-head training reproduced the earlier crashed session *byte-identically*, with all six policy thresholds matching to full float precision, which is the strongest determinism evidence the program has.

Root cause, found after the fact. The channel corruption has a one-line explanation, identified during the Experiment 9 build and stated here because it is the reason Experiment 9 exists. **Qwen3-0.6B-Base** defines no beginning-of-sequence token, so its tokenizer’s `bos_token_id` resolves to the end-of-text token. Experiment 8 seeded every workspace-channel teacher-forcing target and every workspace decode with that identifier, which placed a document boundary in the middle of the sequence. The model did the reasonable thing and began a fresh document, which in this pretraining corpus means a fresh dialogue turn. The causal channel never used that seed and was never affected, which is exactly the observed damage pattern. This is a line-level defect with an architecture-level consequence, and it was invisible to every training-time metric because it lives only in the emission path.

9.10 Experiment 9: Repaired channel, decoupled claims

Experiment 9 was built, unit-tested and preregistered before its run, and this section keeps that order: the design is stated first, exactly as frozen, and the results that follow are judged against it. It responds to the two things Experiment 8 established: a line-level defect that broke the workspace generation channel, and a headline comparison that entangled generation quality with selection quality and then asked both to be demonstrated on a few dozen rows.

Decoupling the claims. Experiment 8 could only ever select among workspace-channel candidates, so a broken channel doomed the selection claim mechanically. Experiment 9 separates three claims onto three surfaces, each independently powered.

- **Claim G, generation.** The repaired workspace channel reaches *parity* with the causal channel on identical rows: zero turn-scaffold emissions, graded accuracy within 0.05 of causal greedy decoding, token F1 at least 0.9 of causal, a causally live read-out (ablation drop at least 0.05) and an open prefix gate. Parity and liveness only. At this scale there is no “beats” claim.
- **Claim S, selection.** Verifier-weighted self-consistency beats plurality self-consistency on the *causal-channel* pool, the same healthy pool the baseline votes over, so channel quality cannot contaminate the comparison. The heads score arbitrary token sequences by teacher-forcing them under the workspace prefix.
- **Claim A, abstention.** Conformally calibrated heads beat mean-log-probability abstention on the risk-coverage curve over the same causal pool.

Channel repair. The answer channel never begins with `bos_token_id`; teacher-forcing predicts the first answer token from the last hard cue token, and decoding starts from that cue. Both channels share one prompt builder: the context loses its trailing response header at encode time and the model appends the header itself as hard cue tokens. The causal channel is then `[context][cue] → answer` and the workspace channel is `[context][prefix][cue] → answer`, so the two differ only by the latent prefix (Figure 6). Position zero remains a hard token, the prompt always ends in hard tokens, and the prefix sits adjacent to the cue where its influence is largest. Context, cue and answer positions all carry the causal mode embedding; only prefix positions carry the synthesis and private ones, which makes the causal channel literally the workspace channel with the prefix removed, and therefore makes the read-out ablation exact rather than approximate. The whole prefix is scaled by a single learned gate $\tanh(\alpha)$ with α initialized at 0.05 and logged at every evaluation boundary; the memory and summary projections move from zero initialization to small random initialization, so the zero now lives in the gate and the diversity and orthogonality gradients are live from the first step instead of provably inert. On anchor steps an auxiliary term $\beta \cdot \text{KL}$ between the

workspace-conditioned answer logits and the detached causal answer logits anchors fluency, with $\beta = 0.05$ and the read-out-liveness gate as its declared canary: too strong a β nulls the mechanism, and the gate is what would catch that. Decode-time banning of scaffold tokens is deliberately *not* added, so the leak gate remains a pure measurement and a failed repair cannot hide behind a filter.

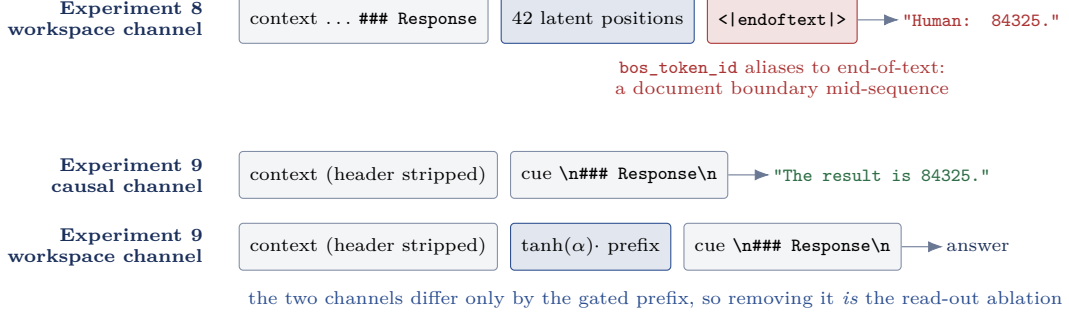


Figure 6: The Experiment 8 defect and the Experiment 9 repair. Experiment 8 seeded every workspace teacher-forcing target and decode with `bos_token_id`, which `Qwen3-0.6B-Base` does not define and whose tokenizer therefore resolves to the end-of-text token, injecting a document boundary in mid-sequence; the model began a fresh document, which in this corpus means a fresh dialogue turn. The causal channel never used that seed and was never damaged, matching the observed pattern exactly. Experiment 9 removes the seed, moves the response header out of the prompt string and into hard cue tokens the model appends itself, and places the latent prefix between context and cue so that the two channels form a matched pair.

Candidate machinery and conformal calibration. The candidate pool is $N = 8$ drawn from the causal channel: one greedy decode plus seven samples at a single temperature of 0.8. Experiment 8’s temperature ladder is dropped because it confounds selection quality with decode quality and, on seed 17, contributed no additional valid candidates. The heads score every candidate by teacher-forcing it under the gated prefix, and the publish score is $\sigma(w \cdot [\text{commit}, \text{risk}, \text{verifier}, \exp(\log p)] + b)$. The fourth feature matters: it nests the log-probability baseline inside the combiner, so a win is a win over something the combiner could have simply copied. Weighted self-consistency clusters candidates by answer signature and weights each cluster by the sum of its members’ publish scores; plain self-consistency is the same rule with unit weights, and tie rates are reported. Policy-head training and calibration both harvest from this same pool configuration, which makes the heads on-policy with respect to the audit.

Calibration uses 256 validation anchors split by episode-id hash into halves A and B of 128. The combiner (w, b) is fitted on split A ; on split B each anchor contributes one score, and the threshold is the k -th smallest with $k = \lfloor (n_B + 1)(1 - c) \rfloor$ at $c = 0.35$, inflated so that realized coverage is at least 0.25 with about 95% probability. If more than 20% of split- B scores tie at the k -th order statistic, a preregistered randomized tie-break applies. There is no midpoint fallback: that is the road that produced seed 29’s coverage of 0.083, and if the combiner cannot fit, the run reports calibration failure.

Lanes, infrastructure and budget. The lanes get one structural last chance. Lane 0’s brief now pools the first half of the context positions and lane 1’s the second half, which makes the inputs asymmetric by construction rather than by penalty; the lane projections are randomly initialized at distinct scales, so the existing squared-cosine orthogonality pressure has a gradient for the first time. The preregistered exit is unconditional: a mean lane-summary cosine of 0.90 or above would be the eighth such failure, and the lanes are then demoted to a single lane and this paper’s lane-diversity sections join routing and fan-in as proposal plus negative evidence. No further lane fixes are authorized. A key-value cache in the decode paths, with cached post-RoPE keys and threaded absolute positions, is what affords an audit of 320 rows instead of 64: all 256 test behavior anchors plus 64 text-to-SQL rows. Difficulty bins are computed from the same eight samples used by every arm (easy is pass@1 above 5/8, medium is [2/8, 5/8], hard is below 2/8), which is label-dependent stratification and is disclosed as such; the headline runs on the medium bin and every bin plus the pooled result is reported. The training canary now exercises the workspace decode path and counts scaffold leaks, which Experiment 8’s canary could not see. The trainable-parameter delta against Experiment 8 is one scalar and a set of re-initializations, under 0.1%.

Gates, pre-mortem and the binding ladder. The battery is 24 gates: six on training and heads, six on Claim G, five on Claim S, five on Claim A and two general. The Claim S headline, verifier-weighted self-consistency beating plurality self-consistency on the medium bin, additionally requires a pooled one-sided McNemar mid- p below 0.05 on the discordant pairs for the replication verdict, is recomputed anchors-only as a sensitivity check against the text-to-SQL rows distorting the bin, and is guarded by a *heads_not_logprob* check requiring the Spearman correlation between publish score and mean log-probability to stay below 0.95 in absolute value. A *selection_headroom_present* gate requires at least 10% of medium-bin rows to have exactly one correct candidate among the eight; if that gate fails, Claim S is declared void for lack of headroom, which is a design result about the difficulty of the rows rather than a failure of the model. Claim A requires audit coverage inside $[0.20, 0.50]$, an AUGRC advantage over the log-probability baseline in at least 90% of 1,000 paired bootstrap resamples per seed, a Clopper–Pearson 95% upper bound on selective risk no worse than 0.45, and the safe-abstention probes. Fifteen predicted failure modes are recorded in advance, each with a detector and a response fixed before the run; among them, an attenuated prefix, a KL anchor strong enough to null the mechanism, a gate that never opens, a threshold landing in a tie mass, a coverage band missed on one seed, numerical drift from the new cache, and a weighted-self-consistency win that turns out to be length bias rather than verification.

The ladder is binding and has five rungs. A full pass on the cores of G, S and A authorizes the matched plain-LoRA control immediately, and nothing else authorizes it. If G passes and S fails with headroom present, selection joins the negative evidence and Claim A stands or falls on its own gates. If G passes and S is void for lack of headroom, the selection question moves to harder rows in the benchmark kit rather than being rerun at this difficulty. If G fails on parity or leak, the latent-workspace generation channel is falsified at this scale even with a repaired interface, the program’s generative claim ends, and head and abstention work continues on the causal channel alone. If A fails on both seeds while G passes, the abstention claim dies and Experiment 9 is reported as the terminal negative result for calibrated selective prediction at 0.6B. There is no rung on which the program is allowed to declare success by narrative.

Outcome. Experiment 9 ran as a single dual-seed session, 4,224 of 4,224 microsteps per seed with zero skipped optimizer updates, followed by fresh-process audits of 320 rows per seed at roughly 19 seconds per row. Both seeds failed the battery at 17 of 24 gates, with six of the seven failures shared, and the shared six state one result twice: every repair in the experiment worked and replicated, and every mechanism the repairs were built to expose measured null or lost to its training-free baseline, and that replicated too. Table 15 carries the headline rows. One session defect is disclosed up front: after both audits and both gate computations had completed and been written to disk, the notebook crashed in a display-only cell that referenced a Experiment 8 record field name, so the pooled selection statistic and the archive bundles were recomputed offline from the on-disk artifacts using the preregistered formulas, and the mid-training metrics logs were fetched from the run’s storage and byte-verified against the training heartbeat before use.

The repairs held. The scaffold leak that defined Experiment 8 is gone: zero leaked rows in 640 audited decodes, and the training canary, which now exercises the workspace decode path, recorded a zero format-marker rate at seventeen of eighteen evaluation boundaries across both seeds, with one transient one-in-eight reading on seed 29 during the local phase, absent thereafter. Channel parity followed: workspace-conditioned greedy decoding matched causal decoding to within 0.013 on seed 17 and exceeded it by 0.003 on seed 29, with token-F1 ratios above one on both. The bos/end-of-text diagnosis of Section 9.9 is thereby confirmed at 320-row scale on two seeds: remove the seeded identifier and the channel damage vanishes entirely. The conformal machinery also did what it was introduced to do. Both seeds fitted non-degenerate four-feature combiners with thresholds taken from the held-out half, tie mass 0.008 and no jitter needed, and realized audit coverage landed at 0.244 and 0.188 against Experiment 8 publish coverages of 0.172 and 0.078. Seed 29’s 0.188 sits 0.013 below the preregistered band and fails its gate; the pre-mortem named a one-seed band miss in advance, and the shortfall is validation-to-test shift, not a return of the degenerate-threshold pathology.

The read-out is inert, and the instruments needed correcting first. Two instrument corrections from a post-session code audit are disclosed ahead of the finding, because both had initially overstated it. First, the shipped ablation did not remove the whole prefix: it sliced away the 34 memory positions

Table 15: Experiment 9 held-out results. Audits are 320 rows per seed: all 256 test behavior anchors plus 64 text-to-SQL rows, decoded fresh from checksummed checkpoints. Red marks preregistered gate failures.

Measurement	Seed 17	Seed 29	Interpretation
Completed microsteps / skipped updates	4,224 / 0	4,224 / 0	Training was stable on both seeds.
Turn-scaffold leak, 320 audited rows	0	0	The register repair held on the emission path.
Causal minus workspace greedy accuracy	+0.013	−0.003	Experiment 8’s thirty-point channel gap is gone.
Workspace / causal token F1 ratio	1.015	1.025	The workspace channel writes marginally cleaner text.
Memory-token ablation (34 of 42 positions)	0.000	−0.022	Removing the latent memory costs nothing.
Prefix gate $\tanh(\alpha)$, full-run range	0.0500–0.0507	0.0498–0.0504	Flat by construction; disqualified as evidence (see text).
Medium-bin exactly-one-correct fraction	0.000	0.000	Claim S void for headroom, as preregistered.
Weighted vs. plurality SC discordant pairs	0 / 0	2 / 0	Two flipped votes in 640 rows; pooled mid- p 0.125.
Conformal coverage, band [0.20, 0.50]	0.244	0.188	The floor held on one seed, missed by 0.013 on the other.
Published answers correct	78 / 78	60 / 60	Perfect selective precision at both operating points.
AUGRC, heads vs. log-probability	0.0637 / 0.0594	0.0654 / 0.0623	The free baseline wins on both seeds.
Mean lane-summary cosine	0.940	0.975	Lane failures eight and nine; demotion executes.
Gates passed	17 / 24	17 / 24	Both seeds fail; the ladder decides, not the narrative.

and kept the 8 gated synthesis tokens in both arms, so the measured 0.000 and −0.022 changes in graded accuracy, with token-F1 changes of 0.0003 and −0.003, bound the memory tokens only. The full prefix is bounded by the stronger instrument, channel parity: the causal arm *is* the workspace arm with the entire prefix removed, and it sits within 0.013 of it on both seeds. Second, the flat gate trajectory (Figure 7) is not evidence that training rejected the prefix, because the gate could not have moved either way: the first operation the frozen model applies to the gated embeddings is a root-mean-square normalization, which is scale-invariant, so the attention-visible content of the prefix is independent of α for any positive value, and the loss gradient on the gate through that dominant branch is approximately zero whether or not the prefix is useful. The channel was open at its initialization, and the decoder ignored its content through attention. A third design fact completes the diagnosis: the fluency anchor’s reference distribution is this same model on the same inputs with the prefix removed, so at convergence of the answer loss the only persistent gradient on the prefix pathway points exactly at changing the logits by nothing. With both channels conditioned on the same full context, the latent was redundant by construction. Experiment 8’s non-null ablation is reinterpreted as the removal of a *corrupting* prefix rather than an informative one, and the mechanism-level conclusion survives the corrections in sharpened form: the model can minimize its loss while ignoring the workspace, and it does. Parity, the one Claim G surface that passed, was achieved by irrelevance.

Selection was void, as its own gate anticipated. The realized audit difficulty is bimodal, not banded: 187 and 189 of 320 rows are easy, with all eight candidates correct and every arm at 1.000; 91 and 98 rows are hard, with any-valid oracles of 0.143 and 0.122; and the medium strata of 42 and 33 rows contain no exactly-one-correct rows at all, so *selection_headroom_present* fails at 0.000 on both seeds and Claim S is void for lack of headroom, the design outcome the gate was written to name. The weighting itself was nearly inert: verifier-weighted and plurality self-consistency produced zero discordant pairs on seed 17 and two on seed 29, two flipped votes in 640 rows, for a pooled McNemar mid- p of 0.125. Report-only signals are recorded without weight: on medium rows the head-argmax arm beat greedy on both seeds, 0.786 against 0.643 and 0.818 against 0.697, with best-of- N log-probability comparable, which is consistent with selection signal existing at difficulty strata this audit barely samples. Per the void rung, the selection question moves to rows with measured headroom rather than being rerun at this difficulty.

Abstention failed its comparison and honored its contract. The preregistered Claim A comparison is lost on both seeds: heads AUGRC 0.0637 and 0.0654 against log-probability 0.0594 and 0.0623, with

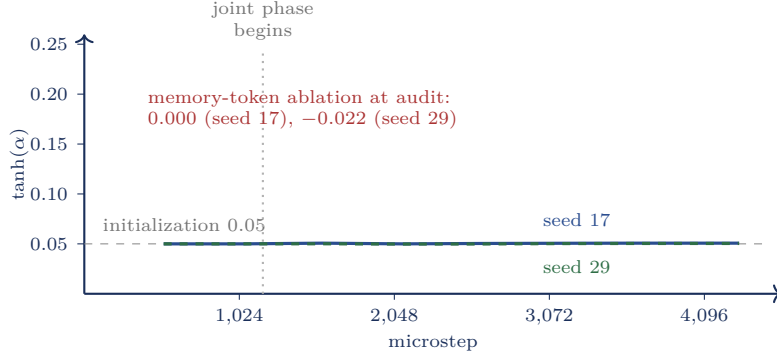


Figure 7: The prefix gate is structurally silent. $\tanh(\alpha)$ at every evaluation boundary of the Experiment 9 run, plotted on the range $[0, 0.26]$; a fully open gate would sit near 1.0, far off scale. The trajectory moves less than 0.0008 on both seeds, but this flatness is forced rather than informative: the frozen model normalizes the gated embeddings before attention, so the content visible to the decoder is independent of α and the gradient on the gate is approximately zero whether or not the prefix helps. The gate is retained as telemetry and disqualified as evidence; the read-out finding rests on channel parity and the memory-token ablation, annotated in red.

paired-bootstrap win fractions of 0.13 and 0.179 against the 0.90 requirement. The guard gates passed, which makes the loss informative rather than degenerate: Spearman correlation between publish score and mean log-probability is 0.830 and 0.821, under the 0.95 ceiling, so the heads are not a re-derivation of likelihood; they are simply not better than it on this distribution. The operating points themselves replicated exactly for the first time in the program: 78 of 78 and 60 of 60 published answers correct, selective-risk upper bounds of 0.038 and 0.049, safe-abstention probes at 1.000, and commit coverage of exactly 0.000 on text-to-SQL on both seeds, the one audited domain where the model is incompetent, with quality 0.078 and 0.109, which is also what drags the pooled quality gate to 0.663 and 0.666 against its 0.70 floor. The shape of the loss is instructive: with 58 to 59 percent of audit rows trivially easy, the log-probability curve is near ceiling on much of its length, and a coarse, near-binary score can win its operating point while losing an area metric. We report the operational property and decline to rename it a victory.

The remaining failures. The verifier head ranked corrupt above clean at 0.944 and 0.947 on teacher-forced features and inverted to -0.233 and -0.297 on freshly generated audit emissions, the program’s cleanest instance yet of train-to-generation distribution shift in a learned judge. The on-policy validity floor was unmet on both seeds: 1,024 emission attempts yielded 2, 5 and 2 valid positives on one seed and 1, 2 and 5 on the other for the numeric, ordering and unit families, against a floor of 16 each, so the heads trained on roughly 25 positives against 1,000 negatives. A post-session code audit reclassified this failure as an implementation defect at least as much as model incapacity: the harvest generated from the training collator’s fixed-length padded prompt surface, roughly 150 padding tokens inserted between prompt and response cue, while the audit encodes prompts unpadded, and on the audit surface the same anchor families reach causal validity of 0.82. The same padded surface feeds calibration, which plausibly contributes to seed 29’s validation-to-test coverage shift. A preregistration-versus-implementation gap is also disclosed here rather than left to a reader: the plan text describes the corresponding gate as enforcing the validity floor, while the shipped gate only requires that head training occurred, so the gate records a pass over a floor that was not met. Lane-summary cosines of 0.940 and 0.975 are lane-diversity failures eight and nine; the unconditional exit executes, the design drops to a single lane, and this paper’s lane sections join routing and fan-in as proposal plus negative evidence.

Decisions taken. The executable preregistration, the notebook inside the sha-verified bundle, enumerates a dead read-out in its stop rung; the prose ladder above, frozen later, words that rung as parity or leak. The older executable document governs, the wording gap is disclosed here, and the stop rung applies. The generative read-out mechanism is retired at this scale, which makes the score zero for three: all mechanisms proposed in Section 2 have now taken preregistered tests and failed them. The matched plain-LoRA control remains unauthorized, nine experiments and zero full passes, and the external benchmarks remain gated behind it. What this experiment adds beyond the retirements is a constraint that any

successor inherits: the failure was not corruption, capacity, or optimization instability, each of which was individually repaired or ruled out along the way, but redundancy. A successor workspace claim must make the latent structurally necessary, for instance by an information asymmetry in which the causal channel does not see everything the workspace sees, so that ignoring the latent costs loss, and it must preregister that necessity check as its first gate.

9.11 Experiment 10: The necessary channel

Experiment 9 ended with a constraint: a successor workspace claim must make the latent structurally necessary, so that ignoring it costs loss, and must preregister that necessity check as its first gate. Experiment 10 is that successor, built and preregistered before its run. On *masked* behavior anchors, the operative premise is moved out of the request the answer path reads: the decoder’s input ids for the answer channel physically lack the operands, the workspace encoder alone sees the full context, and the only route from the premise to the answer is the latent. Training itself uses the same asymmetry on alternating batches, so a model that ignores the latent pays cross-entropy for it during training, not only at audit. The design carries the accumulated instrument corrections forward: the read-out is re-pointed at the evidence encoding rather than the private canvas, the scale-invariant prefix gate is deleted and its telemetry retained without evidentiary weight, the fluency anchor is flipped from the prefix-free reference (measured in Experiment 9 as directed pressure to close the channel) to a distillation teacher, the harvest and calibration surfaces are unpadding (closing Experiment 9’s disclosed padded-surface defect), the validity floor is a real gate rather than a recorded aspiration, LoRA extends to all layers (132.2M trainable against Experiment 9’s 76.6M, disclosed), and anchor difficulty is parameterized (chain length, operand digits, distractors) with the audit pool banded by pass@32 under a frozen estimator whose 224 row ids are fixed before training. Auxiliary instruments are added on both sides of the question: a latent-only linear probe trained to decode the withheld premise, a *trained* gist attribution control (a matched prefix trained on alternating batches with no evidence access), and a within-family shuffled-prefix arm as the primary necessity control. The battery is 22 gates; the binding ladder’s rung 1, a dead channel under structural necessity, is preregistered as terminal for the generative mechanism.

Session record. The first launch crashed at rung 0: both seeds completed all 3,200 microsteps and on-policy collection, then died at the same line inside calibration, where the publish-rule fitter retained Experiment 9’s three-or-four feature guard while the preregistered form selection fits four- and five-feature rules. The two-line fix and a regression test covering the exact call were applied, the bundle rebuilt (only the trainer, its test file and the manifest changed; all data files CRC-identical), and the relaunch resumed from the crashed run’s step-3,200 checkpoints, which carry optimizer, scaler, and python/torch/cuda RNG state. The resume is bit-equivalent to an uninterrupted run: the on-policy collection counts reproduced digit-for-digit (41 and 43 valid emissions of 512, with identical per-family splits). One gate artifact follows from the salvage and is disclosed rather than counted: the canary-cleanliness gate reads the resumed process’s metrics log, which contains only the resume marker, so it records false on both seeds; recomputing the gate’s exact rule on the original training logs gives seven canary records per seed, zero dirty boundaries, zero prompt leaks. The effective battery result is 15 of 22 gates per seed, with seven real failures, identical across seeds.

The necessity result. Every alternative explanation the program has previously needed was ruled out by the run itself. The masking held: zero leaked premise literals in 198 masked audits under a line-anchored scanner shared by trainer and audit. The floor held: with the prefix removed, masked accuracy is exactly 0.000, so the question is only answerable through the channel. The attribution control held: a trained gist prefix with no evidence access also reads 0.000, so nothing generic about trained prefixes explains a success, had one occurred. Unmasked parity held on both seeds, with seed 17’s workspace arm above its causal arm (0.508 against 0.471), so the machinery does not damage the rows it cannot help. Against that instrument, the reading is zero: masked workspace accuracy 0.000 on 99 rows per seed, on both seeds, while the latent-only probe decodes 0.297 and 0.334 of the withheld premise. Experiment 9 showed the latent was ignored when redundant; Experiment 10 shows it is *still* unusable when it is the only source, even though the information demonstrably reaches the latent state. The ladder’s terminal rung executes: the generative workspace mechanism is retired at this scale, now under structural necessity rather than redundancy.

Table 16: Experiment 10 held-out results. Audits are 288 rows per seed (224 banded anchors, of which 99 are masked, plus 64 text-to-SQL); the audit pool was banded by a frozen pass@32 estimator before training. Red marks preregistered gate failures.

Measurement	Seed 17	Seed 29	Interpretation
Completed microsteps / skipped updates	3,200 / 0	3,200 / 0	Training stable; resume bit-exact.
Masked-row leak (99 rows) / causal floor	0 / 0.000	0 / 0.000	The necessity instrument is airtight.
Masked workspace accuracy	0.000	0.000	The only path carried nothing usable.
Trained gist-control accuracy (masked)	0.000	0.000	Not an attribution artifact.
Premise probe accuracy (bar 0.50)	0.297	0.334	The latent holds traces generation cannot spend.
Unmasked parity gap (workspace – causal)	+0.037	0.000	The workspace does no harm; seed 17’s is better.
Verifier margin on generated candidates	−0.277	−0.227	Train-to-generation shift persists in the judge.
On-policy validity floor (16/family)	41 / 512	43 / 512	Numeric and unit generation weak at banded difficulty.
Conformal coverage, band [0.20, 0.55]	0.295	0.267	Experiment 8’s strangle stays fixed; both seeds in band.
Published answers correct	83 / 85	77 / 77	Selective risk UCB ₉₅ 0.072 and 0.038.
Partial AUGRC, heads vs. log-probability	0.0128 / 0.0156	0.0073 / 0.0114	Heads better on <i>both</i> seeds, a program first.
Paired-bootstrap win fraction (bar 0.80)	0.746	0.792	Directionally consistent, short of the bar by 0.054 / 0.008.
Gates passed (of 22; canary artifact excluded)	15	15	Rung 1: terminal for the generative mechanism.

What the near-miss preserves. The calibrated-commitment stack recorded its strongest experiment. Conformal coverage landed inside the band on both seeds, the fitted publish rules were continuous (329 and 372 distinct scores), the operating points published 83 of 85 and 77 of 77 correct answers with selective-risk upper bounds of 0.072 and 0.038, and, for the first time in the program, the heads’ risk-coverage curve beat matched mean-log-probability abstention on both seeds in the preregistered coverage band, at paired-bootstrap win fractions of 0.746 and 0.792 against the 0.80 bar. The win fraction is a one-sided p -value in disguise, and at 288 audit rows per seed the design could only have detected an effect roughly three times the observed size; the claim therefore closes unestablished with a recorded effect direction, an effect size, and a power analysis, and any successor inherits an audit-set requirement in the thousands of rows, not hundreds. The verifier head repeated Experiment 9’s distribution shift, ranking teacher-forced features correctly while inverting on freshly generated candidates (−0.277 and −0.227), and the on-policy validity floor was genuinely unmet at banded difficulty (41 and 43 valid of 512 attempts, concentrated in the abstention and ordering families), which is now a measured property of the generation channel rather than a padded-surface artifact.

The sharpened constraint. Experiment 9’s lesson was necessity; Experiment 10’s is that necessity is not sufficiency. The probe-positive, generation-zero signature that survived every repair in this program is, in retrospect, the published failure mode of answer-only supervision on latent reasoning tokens: CODI’s ablation reproduces it exactly, with GSM8k accuracy collapsing from 43.7 to 24.5 when its hidden-state distillation term is removed while the latent remains present and trained [2], and Coconut documents the same class at the same scale [3]. Every published system that does pass content through a latent interface into a frozen decoder shares one ingredient no HLWM experiment had: a dense, token-level objective through the latent itself, reconstruction or distillation, whose targets cannot be predicted without the latent’s content. This program’s experiments supervised the answer and hoped the latent would become useful; ten experiments say it does not, at this scale, under either redundancy or necessity. A successor preregistration exists that combines the retained necessity instrument with dense latent supervision and per-latent targets; it is a new program with its own gates and is not claimed here.

9.12 A preregistered successor: the dense-supervision channel

The necessity experiment sharpened the constraint on any successor: making the latent the only carrier of required information makes its failure measurable, and it fails; the missing ingredient, shared by every published system that does pass content through a latent interface into a frozen decoder, is a dense token-level objective through the latent itself [2, 3]. A successor instantiation implementing that constraint is built, unit-gated, and preregistered; its on-device preflight has passed on the target hardware, and its run had not completed at the time of writing. We describe it here because it is the architecture of Figure 1 in its current form, and because its design is the direct product of the failure record above. No result from it is claimed in this paper.

Four changes define the successor (Figure 8). First, the workspace produces six autoregressive *continuous thoughts*: the model’s own last hidden state, projected through a small trained map and rescaled to the base embedding distribution, re-enters as the next input embedding, so latents are decoder-legible by construction rather than by hope. Second, every latent position receives direct supervision. A teacher pass conditions on the gold reasoning trace and its per-layer hidden states at the answer position become distillation targets for the student pass, which sees only the masked prompt and its own thoughts; each thought must additionally decode its assigned trace segment through the frozen unembedding, and the thoughts jointly must reconstruct the withheld premise from a standing start. Third, the interface moves from an input-embedding splice to per-layer key–value slots behind post-softmax gates initialized small-positive, so no normalization layer can silence the gradient and no weight-decay term can close the channel unopposed, the two mechanisms behind the earlier inert-gate results. Fourth, specialist capacity is routed deterministically by task family, with a detached probe learning to recover the route: load balance is a property of the data loader, and the collapse mode that retired the learned router cannot recur because nothing about the assignment is learned. Every mechanism ships with a blocking on-device preflight (gradient liveness of each loss group, gate-closed equivalence, decode–train parity, memory and throughput gates), and the audit adds paired controls, within-family shuffled thoughts, compute-matched content-free thoughts, slot ablation, and cross-routed experts, so that a positive masked-transfer reading cannot be manufactured by any single confound the program has previously encountered.

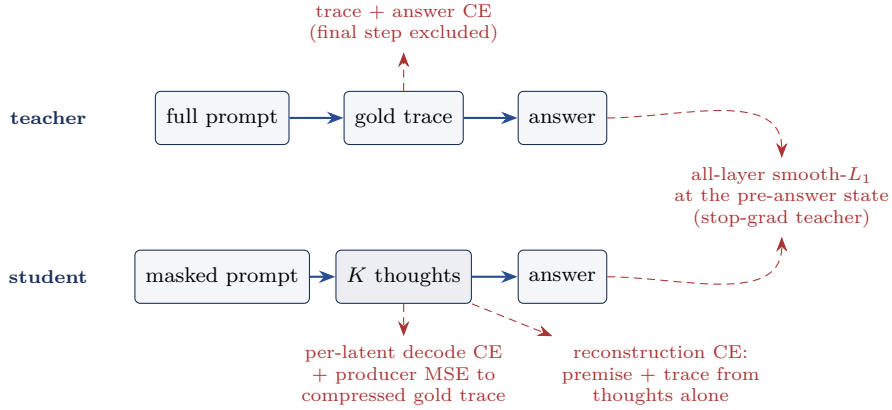


Figure 8: Dense supervision in the successor channel. The teacher pass conditions on the gold trace; the student pass sees only the masked prompt and its own continuous thoughts. Every latent position carries direct losses: per-latent decoding of its trace segment, regression to the compressed gold embedding, standing-start reconstruction of the withheld premise, and all-layer distillation of the teacher’s pre-answer states. Emergence is nowhere load-bearing.

9.13 Scale-up candidate: status and blocking audit

A preregistered protocol exists for a Qwen3-8B-class candidate. It pins Qwen/Qwen3-8B at revision b968826d9c46dd6066d109eabc6255188de91218 and does not transplant the dense weights into a new randomly initialized backbone. The Qwen weights remain resident in 4-bit NF4 form; rank-16 LoRA modules train every attention and MLP projection, while a separately trainable FP32 HLWM sidecar supplies two private lanes, one tied denoising transition, four routed bottleneck experts, a learned latent prefix, a cheap difficulty gate, and publish, risk, and candidate-error heads. Easy supervised rows mask

the latent prefix and train the direct causal route; harder rows train synthesis conditioned on the private workspace. At a sampled corruption time, the same transition executes two recurrent refinements rather than allocating a separately parameterized block. Full reverse-time execution is reserved for generation and verifier training.

The data boundary is stricter than the earlier Reasoning9000 release. Existing episodes contribute only prompts, context, and constraints. Two distinct candidate teachers answer independently; a third model critiques and corrects them while constructing a named hard negative; two further judges must both accept the corrected answer and confirm that the negative is wrong. Every provider response is checkpointed immediately, disagreement fails closed, lineage groups remain split-isolated, and the main 8B launch is disabled until at least 25,000 accepted teacher packets exist. Deterministic arithmetic, conversion, ordering, and missing-evidence anchors are appended separately for execution checks and on-policy labels; they do not substitute for broad independent adjudication. Table 17 summarizes the protocol commitments.

Table 17: Preregistered Qwen3-8B candidate protocol. These are implementation commitments, not observed results.

Component	Committed setting	Failure boundary
Backbone adaptation	Pinned Qwen3-8B; NF4 double quantization; rank-16 LoRA on <code>q/k/v/o/gate/up/down</code> projections.	No claim of a from-scratch model or full-weight conversion.
Native sidecar	Latent width 384; two 16-token lanes; four experts; four diffusion times; two tied refinements; four prefix tokens.	A sidecar that does not improve a compute-matched Qwen control is removed.
Distributed execution	One model trained by two DDP processes on two 15 GB T4 GPUs.	Both devices must pass the real-model forward/backward preflight.
Training phases	Joint causal, denoising, routing, diversity, difficulty, and paired-policy losses; then generated training-anchor policy updates.	Validation and test labels may not enter gradient updates.
Persistence	Atomic checksummed adapter, sidecar, optimizer, scheduler, scaler, per-rank RNG, phase, and exact microstep; time, signal, and stop-file exits.	A checksum, run-signature, data-snapshot, or world-size mismatch blocks resume.
Evaluation	Validation-only publish calibration; untouched test anchors; original Qwen versus adaptive HLWM quality, latency, peak allocation, tokens/s, power, and watt-hours/output.	A failed gate is reported as a failed candidate, not relabeled production-ready.

Kaggle sessions use an 8.5-hour run budget with a 20-minute save buffer, periodic atomic checkpoints, an explicit stop file, and optional synchronization to a private model repository. Numbered checkpoint parts provide a second recovery path when remote synchronization is unavailable. The frozen data manifest carries row counts and SHA-256 hashes for all three splits; the checkpoint stores the matching aggregate fingerprint and refuses a changed snapshot. This protocol reduces repeated manual intervention and artifact loss, but it does not make intermittent free hardware equivalent to a controlled training cluster.

The first narrow capability gate requires validation calibration to contain both correct and incorrect generated candidates, balanced accuracy of at least 0.70, published precision of at least 0.80, held-out semantic accuracy of at least 0.75, no more than a two-point accuracy regression relative to the pinned Qwen control, zero observed prompt leakage, and exercised routing. These thresholds are screening criteria only. A passing result must still be followed by broad independently authored tasks, contamination checks, safety evaluation, repeated seeds, and compute-matched dense and flat-routing controls before any production or frontier-model comparison.

Launch status: on hold. A pre-launch audit found split leakage in the shipped data snapshot (a subset of test prompts verbatim in train, and numeric test items derivable from train items by a constant offset), a degenerate difficulty column that would zero the sidecar’s contribution in every joint batch (reducing the run to plain LoRA finetuning), and a base-model pin inconsistency between the protocol document and the live training source. The candidate remains preregistered, but no 8B result exists and none should be expected until the data snapshot is regenerated and re-audited and the pin is reconciled. Publishing this blocking audit is itself part of the protocol: a failed precondition is reported, not repaired silently.

10. Failure Modes and Decision Rules

The architecture should be simplified to the strongest successful ablation if hierarchy remains decorative, branches collapse, verification fails calibration, or variable routing does not improve the measured quality–compute frontier. This rule has now fired on all three mechanisms. Experiment 6 invoked the expert-count reduction, which Experiment 7 executed and lost; Experiment 8 invoked rung 2 of its own ladder, which retired verified fan-in; Experiment 9’s stop rung retired the latent read-out, and its unconditional lane exit executed on failures eight and nine, demoting the design to a single lane. The rule is only worth stating if it is allowed to remove things, and the record above is what that looks like.

11. Limitations

Honesty about the prototype’s distance from the specification is part of the protocol.

1. **The decisive control has not run.** Every reported comparison is against the *frozen untrained* base, or against training-free baselines drawn from the same trained adapter. The matched control, plain LoRA on the same substrate, data, steps and trainable-parameter budget with no HLWM modules, is the first row of Table 4 and remains the next required experiment. It is also still unauthorized: the preregistered ladder gates it behind a full capability-gate pass, which no experiment has produced. Until it runs, no observed gain can be attributed to the architecture rather than to finetuning at all.
2. **The workspace read-out has never carried usable information.** Through Experiment 7 the synthesis interface was a linear reshape of three pooled vectors into eight positions. Experiment 8 widened it to 42 positions and found a channel that was causally live and harmful at once, which a register defect explained. Experiment 9 repaired the defect and obtained parity, zero scaffold leaks in 640 audited rows and workspace accuracy within 0.013 of causal, and the finding is then bounded twice: ablating the memory positions changes graded accuracy by 0.000 on one seed and improves it by 0.022 on the other, and parity itself bounds the whole prefix, since the causal arm is the workspace arm with the prefix removed. The gate telemetry proved structurally uninformative, because the model’s input normalization makes the visible prefix content independent of the gate value. At prototype scale the latent workspace has never been shown to contribute information to the public answer. The open requirement is structural necessity, not further interface repair (Section 9.10).
3. **Correlated policy labels, partly resolved.** The prototype’s three policy heads were trained on label vectors that are perfectly anti-correlated by construction, so the three-threshold gate collapsed to one effective threshold in six successive runs. Experiment 8 de-correlated the failure events and replaced the triple sweep with a single scalar rule, and the fitted thresholds were non-degenerate on both seeds for the first time. The residual limitation is that the heads still share a trunk and are still supervised by graders in four narrow families, so correlated blind spots remain untested.
4. **Inference cost and audit size.** Through Experiment 8 the prototype’s full reverse schedule re-encoded generously and reused no key–value cache across refinement steps; audits of 64 published rows per seed, of which 33 to 36 carry an authorized grader, are the direct consequence, and every baseline comparison in Section 9.9 rests on that sample size. Experiment 9 added the cache and ran 320-row audits per seed at roughly 19 seconds per row, which is the sample size its comparisons rest on. Measured quality–latency frontiers on cached implementations are still required before any efficiency claim.
5. **The selective-prediction claim failed its preregistered test.** Experiment 8 supported it on one seed of two. Experiment 9 ran the better-powered version, a split-conformal rule against matched mean-log-probability abstention on 320 rows per seed, and lost the risk–coverage comparison on both seeds while publishing 78 of 78 and 60 of 60 correct answers at its operating points. The operational property, coverage-controlled publishing with zero selective errors and total refusal on the model’s incompetent domain, is replicated; the claim that it beats the free baseline is not supported, and we do not advance it.
6. **Narrow evidence base.** Through Experiment 4, all policy evidence came from programmatically verified anchors in four task families. Experiment 5 widened this with execution-verified code episodes

Table 18: Principal mechanism risks with defined responses and falsification conditions. Annotations record where each risk was observed, mitigated, or fixed in the experiments of Section 9; every row in this table fired at least one annotation during the program.

Risk	Response	Reduce the design if . .
Decorative hierarchy	Intervention tests, connected-path diagnostics, and comparison with flat routing.	Fixed top- k matches quality at lower cost.
Root irrelevance or dominance	Output normalization, learned layer scales, anchor constraints, counterfactual interventions, and fixed-share ablations.	The root has no measurable causal value or constraints erase specialist gains.
Routing collapse (observed: Experiments 3–6; load gates passed on one seed in Experiments 5–6, with shrinking margin)	Hard-assignment load balancing, expert-output diversity pressure, and causal intervention gates.	Experts remain homogeneous after matched training; then cut the expert count (invoked after Experiment 6; Experiment 7 ran two experts, the collapse recurred, and the expert graph exited the small-scale claim).
Joint-phase seed variance (observed: Experiments 5–6; persisted through both targeted fixes)	Nonzero expert initialization so diversity pressure is live from the first step (rescued the generation-degeneration mode in Experiment 6); family-balanced head-phase data; generation-time canaries during training; multi-seed gates.	Workspace outcomes remain a seed lottery after targeted fixes; then simplify the workspace (condition met after Experiment 6; reduction experiment preregistered).
Branch collapse (mitigated in Experiment 2) (recurred: Experiments 8–9, cosines 0.945/0.968 and 0.940/0.975)	Joint briefs, isolation, structurally asymmetric context slices, coverage objectives, and outcome-based diversity.	Multiple lanes remain redundant after matched training; then demote to a single lane (exit executed after Experiment 9).
Verifier correlation	Route-diversified lanes, executable graders, adversarial examples, route-overlap statistics, and correlated false-negative tests.	Verification adds confidence but not error discrimination.
Commitment miscalibration (fixed in Experiment 4); operating-point strangulation (observed: Experiment 8, seed 29 at coverage 0.078) (coverage floor delivered in Experiment 9: 0.244/0.188)	On-policy head training, held-out calibration, split-conformal thresholds with an explicit coverage floor, risk-coverage selection, EVI error analysis, and exact-snapshot scoring.	Risk bounds fail at target coverage, coverage cannot be controlled across seeds, or EVI does not predict useful continuation.
Correlated policy labels (observed: Experiments 3–7, six rail-degenerate thresholds) (structurally answered in Experiment 8)	Independent failure events per head, de-correlated supervision, a single scalar publish rule, and threshold non-degeneracy checks.	The three-head gate still calibrates to a single effective threshold.
Latent-prefix register corruption (observed: Experiment 8, both seeds) (repaired in Experiment 9: leak 0/320, parity within 0.013)	Matched-pair channel construction, no sequence-boundary identifiers inside the answer channel, a leak gate that is a measurement rather than a decode-time filter, and a training canary that exercises the emission path.	The workspace channel cannot reach causal parity even with a repaired interface; then the generative claim ends and calibration work continues on the causal channel.
Latent read-out inertness (observed: Experiment 9, both seeds: memory ablation 0.000/−0.022, parity within 0.013)	Structural necessity: information asymmetry between the channels, latent-only auxiliary predictions, gate-trajectory canaries, and exact matched-pair ablations.	The latent remains causally null even when it is the only carrier of required information; then the workspace claim ends at this scale.
Selector without headroom (observed: Experiment 8 headline) (recurred: Experiment 9, medium exactly-one-correct 0.0 on both seeds)	Difficulty-stratified evaluation, an explicit exactly-one-correct headroom gate, and selection tested on the same pool the baseline votes over.	Voting already saturates the pool; then the selection question moves to harder rows rather than being rerun (void declared in Experiment 9).

Table 19: Execution, systems, and protocol risks with defined responses and falsification conditions. These rows concern the scaled design and were not exercised by the small-scale program.

Risk	Response	Reduce the design if...
Activation explosion	Global union/FLOP budget, within-stage route reuse, stop actions, and capacity penalties.	Tail compute cannot be controlled without losing accuracy.
Premature private halt	Paired next-step utility, fixed-depth sweeps, minimum depth, and halt calibration.	Adaptive depth saves cost only by lowering verified task quality.
Route thrashing	Window-level refresh, switch cost, stable tie-breaking, and route-history diagnostics.	Refresh raises switching or tail latency without specialist intervention gain.
Shared-kernel interference	Mode projections, depth-wise anchors, gradient diagnostics, and untied-depth controls.	Tied recurrence underperforms a parameter- and FLOP-matched untied control.
Diffusion inconsistency	Compare with a causal private decoder; evaluate targeted re-noising and hard step caps.	Parallel refinement loses on sequentially constrained outputs.
Rejected-state leakage	Rebuild public frame from committed context plus verified findings only.	Unsupported rejected claims persist into later output.
Systems inefficiency	Grouped kernels, expert parallelism, route reuse, and utilization-aware batching.	Dynamic routing loses its theoretical sparsity advantage in wall-clock measurements.
Barrier stragglers	Per-lane halting, active-lane compaction, bounded windows, and mean/P95 scheduling metrics.	Slow lanes erase useful parallel latency gains.

and a small set of independently adjudicated builder episodes, but no external benchmarks, adversarial evaluations or serving tests exist yet, and the preregistered ladder keeps benchmarks gated. The Reasoning9000 release remains unreviewed and its policy labels remain masked.

7. **Joint-phase training variance.** Experiment 5 produced a near-passing seed and a collapsed seed from the identical configuration, and the collapse, workspace-prefix degeneration with dialogue-format drift, was invisible to every training-time metric. Experiment 6 sharpened this: the targeted initialization fix eliminated the degeneration but not the routing collapse, which reproduced on the same seed under a different initialization. Experiment 7 reproduced it a third time at two experts, and Experiment 8 showed the same seed-dependence in a mechanism with no experts at all, since the two seeds’ publish rules landed at coverage 0.172 and 0.078 from identical training. Experiment 9’s two seeds then agreed almost exactly, sharing six of their seven failed gates, so the variance appears to concentrate in mechanisms operating near their noise floor rather than in the training loop as such. Multi-seed evaluation remains mandatory for every claim.
8. **Lane diversity was never established, and the exit has been taken.** The lane-summary cosine gate passed exactly once in the program, on one seed of Experiment 7 at 0.829, failed on both seeds of Experiment 8 at 0.945 and 0.968, and failed on both seeds of Experiment 9 at 0.940 and 0.975 with structurally asymmetric briefs in place, failures eight and nine. The preregistered unconditional exit executed: the design is single-lane going forward, and this paper’s lane-diversity sections stand as proposal plus negative evidence alongside routing and verified fan-in.

12. Related Work

Hierarchical mixtures of experts introduced coarse-to-fine conditional decomposition [15], while modern sparse MoE systems established large conditional parameter banks and scalable routing [11, 12, 16]. DeepSeekMoE provides a precedent for separating always-active shared capacity from routed capacity, but does not establish that an HLWM root will learn generalist behavior [17]; HLWM treats that as a behavioral claim requiring anchors and interventions. HLWM differs by selecting a variable-depth connected subgraph for each functional stage and private lane, retaining an always-executed root whose behavioral contribution is explicitly tested, and coordinating all selections under a macrocycle budget.

Discrete diffusion defines principled corruption and denoising in categorical spaces [5]. DiffusionGemma, LLaDA, and Block Diffusion establish language-model-scale precedents for discrete or block diffusion and bidirectional canvas refinement [4, 7, 8]. These mechanisms revise tokens; HLWM uses them as the private execution substrate for separately briefed and verified reasoning lanes rather than as the public generation interface.

DiffusionBlocks reframes neural-network depth as denoising time and reports block-wise training for vision transformers, autoregressive language models, and a small recurrent-depth language model [9]. Its

recurrent experiment motivates the single-sampled-time local estimator in Section 6.3. The evidence is nevertheless at small scale, text results are mixed, and increasing the number of blocks can degrade quality. HLWM therefore does not partition its expert DAG or functional stages into independently trained noise ranges: only the tied fast private transition receives the one-pass objective, followed by short joint unrolling for coordination. Entropy-bounded sampler stopping is used strictly as a token-update mask, never as semantic commitment [10].

Tree of Thoughts and self-consistency retain several candidate trajectories through external search or sampling [18, 19]. Continuous latent reasoning demonstrates that intermediate computation need not be rendered as public chain of thought [20]. Weight-tied recurrent depth, adaptive computation time, and Mixture-of-Depths provide precedents for increasing or allocating executed computation without requiring one fixed-depth path [13, 14, 21]. HLWM turns these general principles into a root-preserving multi-rate computation: the same transition drives fast state-isolated lanes, private consolidation, verified global updates, and synthesis while the expert graph and halt policy vary by lane and time. The paper’s claim concerns this measured interaction, not recurrence alone.

Verifier training and post-hoc calibration motivate explicit risk prediction, but shared parameters create correlated blind spots; internal verification is therefore never treated as proof [22, 23, 24].

13. Conclusion

This paper asked whether a hierarchical latent workspace, comprising a root-preserving expert graph, isolated parallel lanes refined by categorical diffusion, and calibrated atomic publication, improves verified answer quality on a small frozen substrate at matched compute. Ten preregistered experiments return a clear answer: not in the proposed form. The value of the record is that every negative is mechanistic rather than bare. Each mechanism failed in a specific, diagnosable, and ultimately reproducible way, each diagnosis forced an instrument correction that survivors of this design space will need, and the one component that repeatedly worked, calibrated selective prediction, is exactly the component whose training signal made it necessary.

Ten preregistered experiments on a 0.6B frozen substrate have answered the question they were built to answer, mostly by removing possibilities. The substrate hosted the dual-mode workspace stably in every experiment, with zero skipped optimizer updates at every scale attempted. The evaluation stack is honest: graders are semantic, abstention is a publishable answer, thresholds are fitted on generated held-out emissions, audits reload checkpoints in fresh processes, and every gate is declared before the run. All three proposed mechanisms have now been retired by that machinery.

The routing mechanism, the part that decides *which expertise to apply*, failed first. A marginal-entropy regularizer was found to make routing probabilities uniform while leaving decisions concentrated, homogenizing the experts it was meant to diversify; replacing it with hard-assignment load balancing produced, on one seed, the first routing in the program to clear the preregistered load gates, while the sibling seed collapsed entirely. With both preregistered variance fixes applied and everything else frozen, the collapsing seed collapsed again, the passing seed’s diversity decayed to the gate line, and the causal-liveness gate failed on ten of ten seed-runs. The reduction rule then ran the design at two experts, the same seed collapsed a third time, and the expert graph left the small-scale claim, at a measured cost of roughly a quarter of probe content accuracy, which showed the experts had been carrying content while failing every causal gate.

The verified-fan-in mechanism failed second, and its failure had two layers. Experiment 8’s selector was near-oracle and still lost by thirty points, because a single mis-specified token identifier had knocked the workspace generation channel into the wrong register, and Experiment 8’s headline was arithmetically unwinnable besides, with two to four points of selection headroom at its operating accuracy. Experiment 9 repaired both: the channel reached parity with causal decoding, zero scaffold leaks in 640 rows, and the comparison moved to the difficulty stratum where selection could in principle matter. The stratum proved empty, no exactly-one-correct rows on either seed, and the weighting flipped two votes in 640 rows. Selection at this scale is void for headroom on honest data, and demonstrating it will require rows this program’s anchor distribution does not contain.

The latent workspace itself, the architecture’s central object, failed last and most cleanly. With the register defect repaired, the finding is bounded twice: ablating the read-out’s memory positions changed

graded accuracy by 0.000 on one seed and improved it by 0.022 on the other, and the causal arm, which is the workspace arm with the whole prefix removed, sat within 0.013 of it on both seeds. A post-session code audit added the structural half of the diagnosis: the frozen decoder’s input normalization makes the visible prefix content independent of the learned gate, and the fluency anchor’s reference is the same model without the prefix, so the training design’s only stable point was a prefix that changes nothing. The latent workspace at this scale is not damaged, expensive, or unstable; it is unnecessary, and the model behaves accordingly. The mechanism-level lesson generalizes beyond this architecture: a bolted-on reasoning structure whose information is redundant with the decoder’s own input will be ignored by a sufficiently capable decoder, and no amount of interface repair changes that.

Experiment 10 then removed the redundancy and asked again. With the premise physically withheld from the answer path, training conducted under that same asymmetry, and an instrument that was airtight on its own gates, zero leaks, an exact 0.000 causal floor, a trained attribution control at 0.000, and parity preserved on unmasked rows, masked accuracy still read 0.000 on both seeds while a linear probe recovered roughly a third of the withheld premise from the latent state. Necessity, the constraint Experiment 9 bequeathed, is therefore necessary but not sufficient: making the latent the only source makes its failure measurable, and it fails. What ten experiments never gave the latent is a dense token-level objective through the latent itself, the one ingredient shared by the published systems that do pass content through learned latent tokens [2, 3], whose ablations reproduce this program’s probe-positive, generation-zero signature exactly. That is the successor’s hypothesis, preregistered elsewhere, and not this paper’s claim.

The calibrated-commitment half of the architecture survived longest, and its trajectory is the most instructive. The split-conformal publication rule delivered controlled coverage where six earlier runs had produced degenerate thresholds and Experiment 8 had strangled at 0.078; its operating points were flawless in Experiment 9, 138 published answers and 138 correct, and again in Experiment 10, 160 published and 158 correct with risk upper bounds at 0.072 and 0.038; it lost its risk–coverage comparison to matched mean-log-probability abstention on both Experiment 9 seeds, and then beat that baseline directionally on both Experiment 10 seeds, at paired-bootstrap win fractions of 0.746 and 0.792 against the preregistered 0.80 bar. The program’s rules do not permit renaming a near-miss a victory, so calibrated selective prediction at small scale closes as unestablished, with a recorded effect direction, an effect size, and the power analysis a successor needs: at 288 audit rows per seed the design could only detect an effect roughly three times the one observed. The machinery works, replicates, and is one adequately powered audit away from an answer either way.

The program therefore ends where its rules put it, and that is the result. Zero of three mechanisms survive their own preregistered tests; the decisive plain-LoRA control was never authorized and remains unrun, so no capability claim of any kind is made; the scale-up candidate remains blocked behind its data audit. What ten experiments support is a method and a record: gates declared before every run, killer baselines inside every audit, failure modes with named detectors, mechanisms retired on schedule rather than rescued by narrative, and a taxonomy of the specific ways a latent-workspace architecture fails at small scale, each with a detector, each measured, several repaired, none rescued. A reader building in this space inherits the constraint that matters, now in its final form: do not ask whether the model uses your mechanism when it does not have to; arrange the world so it has to, and measure that; and when it still does not, supervise the mechanism directly, because ten experiments of arranging the world were not enough to make loss discover it.

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A. Artifact Provenance and Reproducibility

Every experiment in Section ?? maps to an internal artifact tag that identifies its preregistered plan, training checkpoints, and audit records in the project repository. The tags are reproduced here so that any reported number can be traced to its on-disk provenance.

Table 20: Experiment-to-artifact provenance. Tags identify checkpoints, plans, and audit records in the project repository.

Experiment	Artifact tag	Question
Structural diagnostic	legacy-cpu	Do the code paths execute at all? (41k-parameter CPU network; verifies plumbing only.)
Experiment 1: Substrate transplant	v5.2	Can the frozen Qwen3-0.6B substrate host the dual-mode workspace stably?
Experiment 2: Graded evaluation	v5.3	Do semantic graders, safe-abstention publication, and threshold fitting produce honest measurements?
Experiment 3: Publication boundary	v5.4	Does fixing the publication boundary transfer calibration to generated candidates?
Experiment 4: On-policy calibration & replication	v5.5	Do on-policy policy heads separate the model’s own emissions, and does routing survive preregistered causal gates across independent seeds?
Experiment 5: Routing redesign on verified data	v5.6	Does hard-assignment load balancing produce causally live multi-expert routing, on execution-verified multi-domain data, across independent seeds?
Experiment 6: Targeted variance fixes	v5.6.1	Do nonzero expert initialization and family-balanced head-phase data rescue the collapse mode and lift commit accuracy on the same two seeds, everything else frozen?
Experiment 7: The reduction experiment	v5.6.2	At minimum viable expert count (two), does routing become reliable and causally live, with the same seeds, a validity-gated head phase, a generation-time canary and everything else frozen?
Experiment 8: Verified fan-in	v6.0	With the read-out bottleneck opened and the calibrated heads promoted from gate to selector, does the latent workspace beat self-consistency and logit-based selection at a matched candidate budget?
Experiment 9: Repaired channel, decoupled claims	v8.0	With the workspace channel’s register defect repaired, does the channel reach causal parity, and do the calibrated heads out-select and out-abstain their free baselines on the <i>same</i> healthy candidate pool?
Experiment 10: The necessary channel	v9.0	With the premise physically withheld from the causal path, so that ignoring the latent costs loss, does the workspace carry structurally necessary information into generation?